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A parallel finite element solver for parabolic equations on quantum graphs

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Abstract

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Introduction

1 Analytical and numerical aspects of quantum graphs

Quantum graphs have attracted significant interest as a tool for modeling phenomena involving a wide range of physical systems: from wave propagation in photonic crystals and thin waveguides to molecular dynamics and signal transmission in social, financial networks and neural architectures in the context of neurosciences. One can find a collection of papers on quantum graphs in [6]. Together with their potential interdisciplinarity, they gained popularity also because they naturally arise as reduced-order models.

Here we provide the essential theoretical background on quantum graphs needed for the computational development of this project. We emphasize that the material covered here is by no means exhaustive; a comprehensive introduction to quantum graphs can be found in [4]. Therefore, the purpose of this section is to introduce the reader to the class of mathematical models known as *quantum graphs*, first from an analytical perspective and then by describing the numerical methods used to study them.

1.1 Basic definitions

Let us begin with an intuitive definition. A *quantum graph* is a collection of one-dimensional intervals connected at their endpoints to form a network. The term “quantum” refers to the fact that the graph is equipped with a Hamiltonian operator that acts on the functions supported in these intervals, coupled by suitable boundary conditions at the vertices [3]. We already provided the main idea behind quantum graphs, but for a more rigorous formulation, we need the following Definitions:

Definition 1.1 (directed graphs). A **directed graph** is a pair of sets $\Gamma = (\mathcal{V}, \mathcal{A})$, where:

- $\mathcal{V} = \{v_i\}_{i=1}^N$ is a set of **vertices** (or nodes);
- $\mathcal{A} \subseteq \mathcal{V} \times \mathcal{V}$ is a set of ordered pairs of vertices, called **arcs**.
- Given an arc $(v_i, v_j) \in \mathcal{A}$, it is said to be outgoing from v_i and incoming to v_j . Two nodes v_i and v_j are said to be adjacent if $(v_i, v_j) \in \mathcal{A}$.

Definition 1.2 (undirected graphs). An **undirected graph** is a pair of sets $\Gamma = (\mathcal{V}, \mathcal{E})$, where:

- $\mathcal{V} = \{v_i\}_{i=1}^N$ is a set of **vertices** (or nodes);
- $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is a set of unordered pairs of vertices, called **edges**.
- An edge connecting two vertices v_i, v_j is written $\{v_i, v_j\}$ with $i < j$. Two nodes v_i and v_j are said to be adjacent if $\{v_i, v_j\} \in \mathcal{E}$.

Definition 1.3 (combinatorial and metric graphs). *Definition 1.1 and Definition 1.2 present arcs and edges as binary relations between pairs of vertices. Two nodes are either linked together or not: we don't know anything about the "strength" of the connection. For this reason, the graphs that we presented are called **combinatorial graphs**. One can choose a measure for the length of such connections. A combinatorial graph endowed with geometry is a **metric graph**.*

Now we present some useful matrices used for the description of combinatorial (and metric) graphs.

Definition 1.4 (adjacency matrix). *The **adjacency matrix** of a graph Γ with N vertices is the symmetric $N \times N$ matrix $A \in \{0, 1\}^{N \times N}$, defined by*

$$A_{ij} = \begin{cases} 1 & \text{if vertices } v_i, v_j \text{ are adjacent,} \\ 0 & \text{otherwise.} \end{cases}$$

Definition 1.5 (incidence matrix). *The **incidence matrix** of a directed graph $\Gamma = (\mathcal{V}, \mathcal{A})$ with N vertices and M arcs is the $N \times M$ matrix E , whose column corresponding to arc (v_i, v_j) has exactly two nonzero entries: $+1$ in row i and -1 in row j . The same holds for an undirected graph with edges $\{v_i, v_j\}$.*

Definition 1.6 (combinatorial Laplacian). *The **combinatorial Laplacian** of Γ is the $N \times N$ symmetric positive semidefinite singular matrix*

$$L_\Gamma = EE^\top.$$

For the analysis of the properties of the combinatorial Laplacian, we refer to Appendix A.

Definition 1.7 (natural coordinates). *Each edge e of a (undirected) metric graph can be associated to a finite interval $(0, \ell_e) \subset \mathbb{R}$, with **natural coordinate** s_e , so that s_e gives the position of a point along edge e . Note that assigning a coordinate requires a choice of direction: we take the increasing direction of s_e to coincide with the orientation fixed when defining the incidence matrix E . For $e = \{v_i, v_j\}$, s_e will therefore increase in the direction that goes from v_i to v_j , with $i < j$. The same definition holds for natural coordinates on arcs of a directed graph, with increasing direction coinciding with the one of the arc itself.*

The definitions of metric graphs and natural coordinates are very important to introduce functional spaces over Γ .

Definition 1.8 ($L^2(\Gamma)$). Let $\Gamma = (\mathcal{V}, \mathcal{E})$ be a metric graph. For each edge $e \in \mathcal{E}$, let

$$L^2(e) := \left\{ f: e \rightarrow \mathbb{R} \mid \int_e |f|^2 ds < \infty \right\}.$$

The space of square-integrable functions on Γ is defined as

$$L^2(\Gamma) := \bigoplus_{e \in \mathcal{E}} L^2(e).$$

In other terms, $f \in L^2(\Gamma)$ if and only if

$$\|f\|_{L^2(\Gamma)}^2 = \sum_{e \in \mathcal{E}} \|f\|_{L^2(e)}^2 < \infty.$$

Definition 1.9 ($H^1(\Gamma)$). For each edge $e \in \mathcal{E}$, let

$$H^1(e) := \left\{ f \in L^2(e) \mid \int_e |f'(s)|^2 ds < \infty \right\}.$$

One definition of $H^1(\Gamma)$ is:

$$H^1(\Gamma) := \left(\bigoplus_{e \in \mathcal{E}} H^1(e) \right) \cap C^0(\Gamma),$$

where $C^0(\Gamma)$ is the space of continuous functions on Γ .

In other terms, $f \in H^1(\Gamma)$ if and only if f is continuous on Γ and

$$\|f\|_{H^1(\Gamma)}^2 = \sum_{e \in \mathcal{E}} \|f\|_{H^1(e)}^2 < \infty.$$

The idea behind the definition of a quantum graph is that edges may be characterised not only by constant coefficients, but also by variable ones following physical laws, possibly described by differential equations. In particular, vertices can interact through such laws. To formalise this, we assign to the metric graph Γ an additional structure represented by a differential operator $\mathcal{H}$, called the *Hamiltonian*. In particular:

$$\mathcal{H} u(s) := -\frac{d^2 u(s)}{ds^2} + V(s) u(s), \quad (1)$$

where $V(s) \in L^\infty(\Gamma)$ is a given potential and derivatives are taken along the edge coordinate s . To pose a well-defined differential problem on Γ , we must supplement $\mathcal{H}$ with a set of vertex conditions $\mathcal{C}$.

Definition 1.10 (quantum graphs). A quantum graph is a triple $(\Gamma, \mathcal{H}, \mathcal{C})$, where $\Gamma = (\mathcal{V}, \mathcal{E})$ is a metric graph, $\mathcal{H}$ is a differential operator acting edgewise on it and $\mathcal{C}$ is a set of vertex conditions imposed at $\mathcal{V}$.

Examples of boundary conditions are the so called *Neumann-Kirchhoff conditions*. The remainder of this report is devoted to the investigation of differential problems on

quantum graphs. In particular, we will notice that also more complicated operators acting on Γ can be considered, as the one generating parabolic problems.

1.2 Finite element discretization

The classical finite element method can be employed to study quantum graphs. Let $\Gamma = (\mathcal{V}, \mathcal{E})$ be a metric graph, where each edge $e \in \mathcal{E}$ is identified with an interval $(0, \ell_e)$. For an edge $e = \{v_p, v_q\}$, we fix an orientation so that $s_e = 0$ corresponds to v_p and $s_e = \ell_e$ to v_q . On each edge e , we introduce a uniform partition into n_e subintervals of length $h_e = \ell_e/n_e$. This yields the nodes

$$\{s_j^e\}_{j=0}^{n_e}, \quad s_j^e = jh_e,$$

with $s_0^e = 0$ and $s_{n_e}^e = \ell_e$, corresponding to the vertices v_p and v_q , respectively. The interior nodes are $\{s_j^e\}_{j=1}^{n_e-1}$. This construction induces a new graph, which we refer to as the *extended graph*. Its vertex set consists of the original vertices $\mathcal{V}$ together with all the interior nodes introduced by the discretization, while its edge set is given by the segments connecting consecutive nodes along each original edge. On this extended graph, we construct the standard hat basis functions $\{\phi_j^e\}_{j=1}^{n_e-1}$, associated with the interior nodes of each edge, together with the functions $\{\phi_v\}_{v \in \mathcal{V}}$, related to the original vertices. All these functions are piecewise linear and globally continuous on Γ . More precisely, each function ϕ_j^e is supported on the two elements adjacent to the node s_j^e , takes the value 1 at s_j^e , and vanishes at all other nodes. Similarly, for each vertex $v \in \mathcal{V}$, the function ϕ_v takes the value 1 at v and decreases linearly to zero along each edge incident to v , vanishing at the neighboring nodes. In practice, we first construct, on each edge, the usual hat functions of the standard finite element method (yielding $n_e - 1$ degrees of freedom per edge). We then enrich this construction by adding basis functions associated with the vertices, thereby incorporating the topological structure of the graph. In this way, the classical finite element basis is naturally extended to the setting of (quantum) graphs. We define the local finite element space on each edge as

$$V_h^e := \{u_h \in H^1(e) : u_h \text{ is piecewise linear with respect to the partition}\}.$$

The global finite element space is then given by

$$V_h(\Gamma) := \bigoplus_{e \in \mathcal{E}} V_h(e)$$

which is a finite-dimensional subspace of $H^1(\Gamma)$, in which we'll search approximate solutions to the weak formulation of differential problems. Any function $u_h \in V_h(\Gamma)$ can be written as a linear combination of the basis functions:

$$u_h(s) = \sum_{e \in \mathcal{E}} \sum_{j=1}^{n_e-1} \alpha_j^e \phi_j^e(s) + \sum_{v \in \mathcal{V}} \beta_v \phi_v(s). \quad (2)$$

One observation that we can make is that it is convenient to order the vertices by following the original edge structure of the graph. More precisely, for each edge, the vertices introduced by the discretization (i.e., the interior nodes along that edge) are numbered consecutively. After all such nodes have been listed for every edge, the original vertices of the graph are numbered last. In this way, the

vertices corresponding to the subdivided edges (the subdomain nodes) appear in contiguous blocks within linear algebra structures, while the original graph vertices (the separators) are grouped together at the end of these. This is showed in the following Figure.

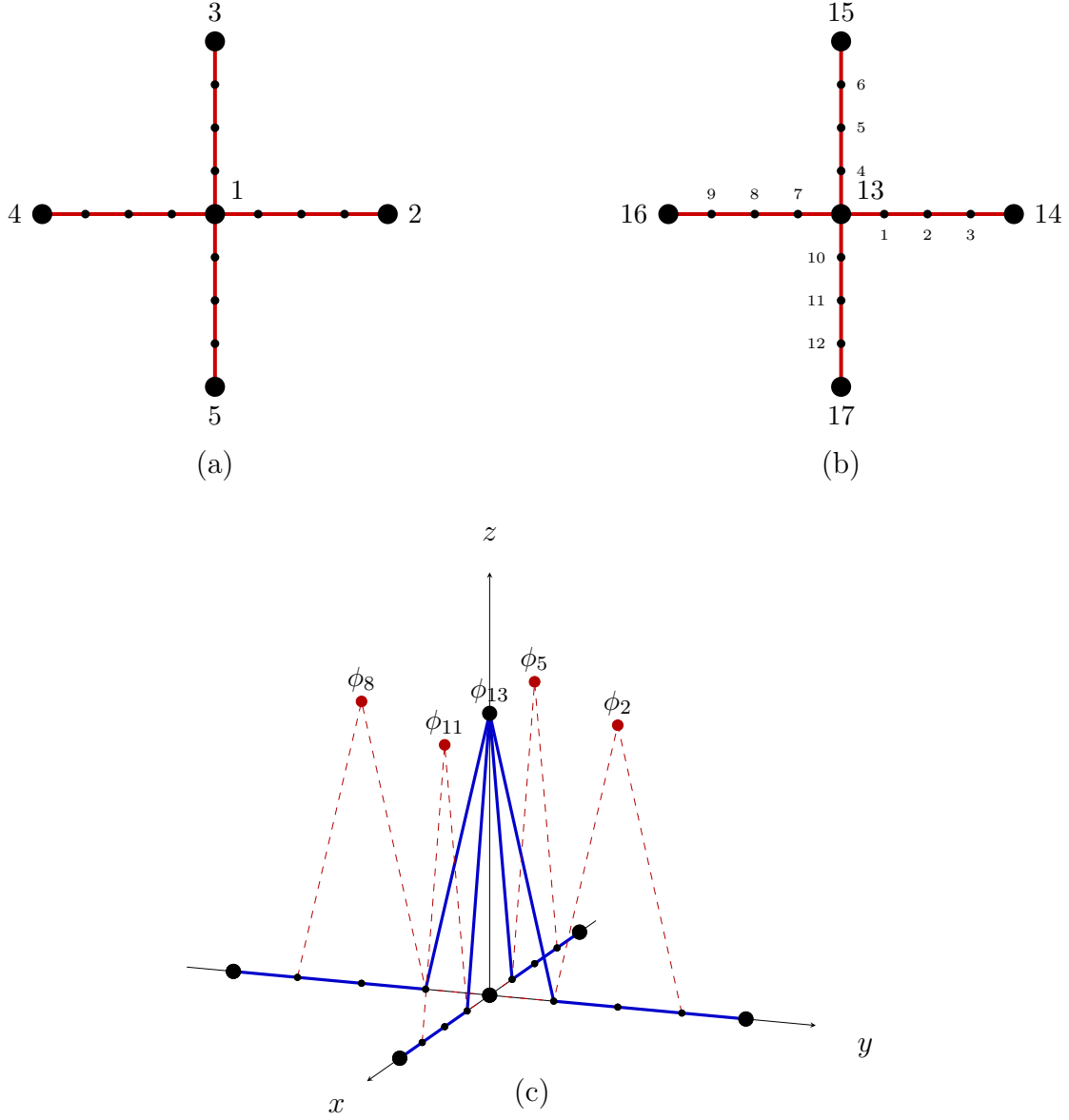


Figure. (a) A star-shaped graph $\Gamma = (\mathcal{V}, \mathcal{E})$ with five vertices ($\mathcal{V} = \{v_1, v_2, v_3, v_4, v_5\}$) and four edges ($\mathcal{E} = \{\{v_1, v_2\}, \{v_1, v_3\}, \{v_1, v_4\}, \{v_1, v_5\}\}$), equipped with a uniform mesh consisting of four elements per edge. The edges incident to the central vertex v_1 are highlighted in red. (b) The same star graph with global node numbering: interior discretisation nodes are listed first, edge by edge (right: 1–3; top: 4–6; left: 7–9; bottom: 10–12, each numbered from centre outward; this orientation coincides with the directionality chosen for the construction of the incidence matrix), followed by the true graph vertices (centre: 13; right: 14; top: 15; left: 16; bottom: 17). (c) The graph Γ together with its finite-element basis. In blue, the vertex basis function ϕ_{13} ; in dashed red, four representative hat basis functions, one per arm.

2 Parabolic problems on metric graphs

A comprehensive analysis on graph-based parabolic problems can be found in [5].

2.1 Problem formulation

We consider quantum graphs with parabolic Hamiltonians. Let $\Gamma = (\mathcal{V}, \mathcal{E})$ be a metric graph, $V \in L^\infty(\Gamma)$ and $f : \Gamma \rightarrow \mathbb{R}$ be given. Let $\mathbb{H}$ denote the elliptic Hamiltonian of the quantum graph $(\Gamma, \mathbb{H}, \mathcal{C}_{NK})$, with underlying Neumann-Kirchhoff conditions at each vertex in $\mathcal{V}$ and global continuity constraints. The parabolic Hamiltonian acts on functions defined on Γ as follows:

$$\mathbb{H} u(s, t) = \frac{\partial u(s, t)}{\partial t} + \mathcal{H} u(s, t) = \frac{\partial u(s, t)}{\partial t} - \frac{\partial^2 u(s, t)}{\partial s^2} + V(s) u(s, t),$$

where $\mathcal{H}$ is the elliptic Hamiltonian introduced with (Eq. 1).

Let us consider the differential problem

$$\begin{cases} \mathbb{H} u = f & \text{in } \Gamma \times (0, T) \\ \sum_{e \in \mathcal{E}_{v_p}} \left[\frac{\partial u_e}{\partial \mathbf{n}_e} \right]_v = 0, & \forall v \in \mathcal{V}, \forall t \in (0, T) \end{cases} \quad (3)$$

where dependence on the variables (s, t) have been omitted in the first equation. This problem can be expressed in weak form by means of a standard procedure reported in Appendix B, yielding

$$\sum_{e \in \mathcal{E}} \int_e \frac{\partial u}{\partial t} w \, ds + \sum_{e \in \mathcal{E}} \int_e \frac{\partial u}{\partial s} \frac{\partial w}{\partial s} \, ds + \sum_{e \in \mathcal{E}} \int_e V u w \, ds = \sum_{e \in \mathcal{E}} \int_e f w \, ds, \forall w \in H^1(\Gamma). \quad (4)$$

Defining

$$a(u, w) = \sum_{e \in \mathcal{E}} \int_e \left[\frac{\partial u}{\partial s} \frac{\partial w}{\partial s} + V u w \right] ds, \quad F(w) = \sum_{e \in \mathcal{E}} \int_e f w \, ds,$$

the weak formulation reads:

Find $u \in H^1[\Gamma \times (0, T)]$, globally continuous on Γ such that

$$\sum_{e \in \mathcal{E}} \int_e \frac{\partial u}{\partial t} w \, ds + a(u, w) = F(w) \quad \forall w \in H^1(\Gamma). \quad (5)$$

The discrete version of the problem can be obtained exploiting a standard Galerkin procedure. Let $V_h \subset H^1(\Gamma)$ be the space spanned by the basis functions $\{\phi_j^e\}_{j=1}^{n_e-1}$ (interior node hat functions on each edge) and $\{\phi_v\}_{v \in \mathcal{V}}$ (vertex basis functions), as defined in SECTION 1.2. The discrete solution is written as in Eq.(2) and substituted into the the weak formulation (5), whose terms are therefore restricted to the subspace V_h . We first test on an interior node hat function of an arbitrary edge. The time term gives:

$$\sum_{e \in \mathcal{E}} \int_e \frac{\partial u_h}{\partial t} \phi_k^{e'} \, ds = \sum_{j=1}^{n_{e'}-1} \dot{\alpha}_j^{e'} \underbrace{\int_{e'} \phi_j^{e'} \phi_k^{e'} \, ds}_{P_{kj}^{e'}} + \sum_{v \in \mathcal{V}} \dot{\beta}_v \underbrace{\int_{e'} \phi_v \phi_k^{e'} \, ds}_{R_{kv}^{e'}}$$

where only the $e = e'$ term survives due to the local support of $\phi_k^{e'}$ and $R_{kv}^{e'}$ is nonzero only for the (at most two) vertices incident to e' . The stiffness term gives:

$$\begin{aligned} a(u_h, \phi_k^{e'}) &= \sum_{j=1}^{n_{e'}-1} \alpha_j^{e'} \underbrace{\int_{e'} \left[\frac{\partial \phi_j^{e'}}{\partial s} \frac{\partial \phi_k^{e'}}{\partial s} + V_{e'} \phi_j^{e'} \phi_k^{e'} \right] ds}_{(H_{11})_{kj}^{e'}} + \\ &+ \sum_{v \in \mathcal{V}} \beta_v \underbrace{\int_{e'} \left[\frac{\partial \phi_v}{\partial s} \frac{\partial \phi_k^{e'}}{\partial s} + V_{e'} \phi_v \phi_k^{e'} \right] ds}_{(H_{12})_{kv}^{e'}}. \end{aligned}$$

The forcing term gives:

$$F(\phi_k^{e'}) = \int_{e'} f_{e'} \phi_k^{e'} ds.$$

We then test on an arbitrary vertex basis function. The time term gives:

$$\sum_{e \in \mathcal{E}} \int_e \frac{\partial u_h}{\partial t} \phi_{v'} ds = \sum_{e \in \mathcal{E}} \sum_{j=1}^{n_e-1} \dot{\alpha}_j^e \underbrace{\int_e \phi_j^e \phi_{v'} ds}_{R_{v'j}^e} + \sum_{v \in \mathcal{V}} \dot{\beta}_v \underbrace{\sum_{e \in \mathcal{E}} \int_e \phi_v \phi_{v'} ds}_{Q_{v'v}},$$

where $Q_{vv'}$ is nonzero only when v and v' are the same. The stiffness term gives:

$$\begin{aligned} a(u_h, \phi_{v'}) &= \sum_{e \in \mathcal{E}} \sum_{j=1}^{n_e-1} \alpha_j^e \underbrace{\int_e \left[\frac{\partial \phi_j^e}{\partial s} \frac{\partial \phi_{v'}}{\partial s} + V_e \phi_j^e \phi_{v'} \right] ds}_{(H_{21})_{v'j}^e} + \\ &+ \sum_{v \in \mathcal{V}} \beta_v \underbrace{\left[\sum_{e \in \mathcal{E}} \int_e \left(\frac{\partial \phi_v}{\partial s} \frac{\partial \phi_{v'}}{\partial s} + V_e \phi_v \phi_{v'} \right) ds \right]}_{(H_{22})_{v'v}}, \end{aligned}$$

while the forcing term

$$F(\phi_{v'}) = \sum_{e \in \mathcal{E}} \int_e f_e \phi_{v'} ds.$$

2.2 Matrix formulation

Collecting all unknowns into $\mathbf{u} = [\mathbf{u}_{\mathcal{E}}, \mathbf{u}_{\mathcal{V}}]^\top$, where $\mathbf{u}_{\mathcal{E}} = \left\{ \{\alpha_j^e\}_{j=1}^{n_e-1} \right\}_{e \in \mathcal{E}}$ and $\mathbf{u}_{\mathcal{V}} = \{\beta_v\}_{v \in \mathcal{V}}$, the Galerkin system is

$$\mathbf{M} \begin{bmatrix} \dot{\mathbf{u}}_{\mathcal{E}}(t) \\ \dot{\mathbf{u}}_{\mathcal{V}}(t) \end{bmatrix} + \mathbf{H} \begin{bmatrix} \mathbf{u}_{\mathcal{E}}(t) \\ \mathbf{u}_{\mathcal{V}}(t) \end{bmatrix} = \begin{bmatrix} \mathbf{f}_{\mathcal{E}}(t) \\ \mathbf{f}_{\mathcal{V}}(t) \end{bmatrix}, \quad (6)$$

with

$$\mathbf{M} = \begin{bmatrix} \mathbf{P} & \mathbf{R} \\ \mathbf{R}^\top & \mathbf{Q} \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{21} & \mathbf{H}_{22} \end{bmatrix}, \quad (7)$$

where the blocks' elements are defined as follows:

- $P_{kj}^e = \int_e \phi_j^e \phi_k^e ds$;

- $R_{kv}^e = \int_e \phi_v \phi_k^e ds;$
- $Q_{v'v} = \sum_{e \in \mathcal{E}} \int_e \phi_v \phi_{v'} ds;$
- $(H_{11})_{kj}^e = \int_e \left[\frac{\partial \phi_j^e}{\partial s} \frac{\partial \phi_k^e}{\partial s} + V_e \phi_j^e \phi_k^e \right] ds;$
- $(H_{12})_{vj}^e = \int_e \left[\frac{\partial \phi_j^e}{\partial s} \frac{\partial \phi_v}{\partial s} + V_e \phi_j^e \phi_v \right] ds;$
- $(H_{22})_{v'v} = \sum_{e \in \mathcal{E}} \int_e \left[\frac{\partial \phi_v}{\partial s} \frac{\partial \phi_{v'}}{\partial s} + V_e \phi_v \phi_{v'} \right] ds.$

Theorem 2.1. *The mass matrix $\mathbf{M}$ and the discrete Hamiltonian $\mathbf{H}$ have the following block structures:*

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_{11} & \mathbf{H}_{12} \\ \mathbf{H}_{12}^\top & \mathbf{H}_{22} \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{G} \end{bmatrix}}_{\mathbf{L}} + \underbrace{\begin{bmatrix} \mathbf{V} & \mathbf{C} \\ \mathbf{C}^\top & \mathbf{D} \end{bmatrix}}_{\mathbf{M}_v},$$

$$\mathbf{M} = \begin{bmatrix} \mathbf{P} & \mathbf{R} \\ \mathbf{R}^\top & \mathbf{Q} \end{bmatrix},$$

where the entries of each block are defined as follows.

In particular:

1. $\mathbf{H}$ is symmetric and positive definite; $\mathbf{M}$ is symmetric positive definite.
2. $\mathbf{A}$ and $\mathbf{V}$ are block diagonal matrices, with one block per edge $e \in \mathcal{E}$, of size $(n_e - 1) \times (n_e - 1)$. Similarly, $\mathbf{P}$ is block diagonal with one block per edge.
3. $\mathbf{G}$ and $\mathbf{D}$ are diagonal matrices of size $N \times N$, being $N = |\mathcal{V}|$. Similarly, $\mathbf{Q}$ is diagonal.
4. $\mathbf{L}$, called the *stiffness matrix*, is the Laplace matrix of the extended graph. $\mathbf{M}_V$, called the *generalized mass matrix*, encodes the potential V . The full discrete Hamiltonian satisfies $\mathbf{H} = \mathbf{L} + \mathbf{M}_v$.
5. The entries of each block of $\mathbf{H}$ are:

$$A_{jk} = \int_e \frac{\partial \phi_j^e}{\partial s} \frac{\partial \phi_k^e}{\partial s} ds = \begin{cases} 2/h_e & \text{if } j = k, \\ -1/h_e & \text{if } |j - k| = 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$B_{vk} = \int_e \frac{\partial \phi_v}{\partial s} \frac{\partial \phi_k^e}{\partial s} ds = \begin{cases} -1/h_e & \text{if } e \in \mathcal{E}_v \cap \mathcal{E}_k \neq \emptyset, \\ 0 & \text{otherwise,} \end{cases}$$

$$G_{vv} = \int_{\mathcal{W}_v} \frac{\partial \phi_v}{\partial s} \frac{\partial \phi_v}{\partial s} ds = \sum_{e \in \mathcal{E}_v} \frac{1}{h_e},$$

$$V_{jk} = \int_e V(s) \phi_j^e(s) \phi_k^e(s) ds, \quad C_{vk} = \int_e V(s) \phi_v(s) \phi_k^e(s) ds, \quad D_{vv} = \int_{\mathcal{W}_v} V(s) \phi_v^2(s) ds.$$

6. The entries of each block of $\mathbf{M}$ are:

$$P_{jk}^e = \int_e \phi_j^e \phi_k^e ds = \begin{cases} 2h_e/6 & \text{if } j = k, \\ h_e/6 & \text{if } |j - k| = 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$R_{kv}^e = \int_e \phi_v \phi_k^e ds = \begin{cases} h_e/6 & \text{if } e \in \mathcal{E}_v \text{ and } k \text{ is the interior node adjacent to } v, \\ 0 & \text{otherwise,} \end{cases}$$

$$Q_{v'v} = \sum_{e \in \mathcal{E}} \int_e \phi_v \phi_{v'} ds = \begin{cases} \sum_{e \in \mathcal{E}_v} \frac{h_e}{3} & \text{if } v' = v, \\ 0 & \text{otherwise.} \end{cases}$$

The diagonal structure of $\mathbf{Q}$ follows from the fact that vertex basis functions ϕ_v and $\phi_{v'}$ have disjoint support whenever $v \neq v'$, since each ϕ_v is supported only on edges incident to v .

2.3 Fully discrete system

Discretizing the semi-discrete system in time by the θ -method with uniform step $\Delta t = T/N_T$ and setting $\mathbf{u}^n \approx \mathbf{u}(t^n)$, one obtains

$$\mathbf{M} \frac{\mathbf{u}^n - \mathbf{u}^{n-1}}{\Delta t} + \mathbf{H}(\theta \mathbf{u}^n + (1 - \theta) \mathbf{u}^{n-1}) = \theta \mathbf{f}^n + (1 - \theta) \mathbf{f}^{n-1},$$

which rearranges to

$$\mathbf{M}_\theta \mathbf{u}^n = (\mathbf{M} - (1 - \theta) \Delta t \mathbf{H}) \mathbf{u}^{n-1} + \Delta t (\theta \mathbf{f}^n + (1 - \theta) \mathbf{f}^{n-1}), \quad (8)$$

where

$$\mathbf{M}_\theta = \mathbf{M} + \theta \Delta t \mathbf{H} = \begin{bmatrix} \mathbf{P} + \theta \Delta t \mathbf{H}_{11} & \mathbf{R} + \theta \Delta t \mathbf{H}_{12} \\ \mathbf{R}^\top + \theta \Delta t \mathbf{H}_{21} & \mathbf{Q} + \theta \Delta t \mathbf{H}_{22} \end{bmatrix}.$$

The choice $\theta = 1$ recovers the backward Euler method, while $\theta = \frac{1}{2}$ gives the Crank–Nicolson scheme.

Since $a(\cdot, \cdot)$ is symmetric and coercive on $H^1(\Gamma)$ and $\mathbf{M}$ is symmetric positive definite, $\mathbf{M}_\theta$ is symmetric positive definite for every $\Delta t > 0$ and $\theta \in (0, 1]$; hence the linear system admits a unique solution at each time step.

3 Numerical Benchmarks

This section presents the numerical benchmarks developed to validate the C++ implementation described in Chapter 5. The tests are organized as a progressive validation path, moving from basic elliptic checks to genuinely parabolic phenomena. Following [1], we begin with three elliptic test cases, denoted E1–E3, which serve as reference benchmarks for the parabolic extension. Test E1 verifies the correctness of the assembly procedure and the basic control flow by prescribing a constant exact solution. Test E2 assesses the enforcement of mixed Dirichlet–Neumann–Kirchhoff boundary conditions through a linear exact solution. Test E3 investigates the spatial convergence of the solver by adopting an exact solution that does not belong to the finite element space (a sinusoidal solution). We then introduce three parabolic benchmarks. Test 1 considers a radially decaying solution on a star-shaped graph, where both diffusion and the reaction term contribute to the time evolution of the solution. Test 2 assesses temporal convergence, verifying that the Backward Euler scheme achieves first-order accuracy and the Crank–Nicolson scheme achieves second-order accuracy in time. Test 3 prescribes a spatially localized initial condition and examines the qualitative propagation of the solution across the graph topology.

3.1 Elliptic Test Cases

We begin with three elliptic benchmark problems on a star-shaped metric graph Γ with five vertices and four edges. In particular, $\Gamma = (\mathcal{V}, \mathcal{E})$, with $\mathcal{V} = \{v_0 = (0, 0), v_1 = (1, 0), v_2 = (0, 1), v_3 = (-1, 0), v_4 = (0, -1)\}$ and $\mathcal{E} = \{\{v_0, v_1\}, \{v_0, v_2\}, \{v_0, v_3\}, \{v_0, v_4\}\}$. Thus, Γ consists of four unit-length edges meeting at the central vertex v_0 , like the one represented in Section 1.2.

A key observation is the following: these benchmarks, while representing elliptic cases, are used to validate a parabolic pipeline, so they will be interpreted in this setting.

3.1.1 Test E1: constant function

The first elliptic test considers a constant exact solution on the star graph:

$$u_{\text{exact}} = 5,$$

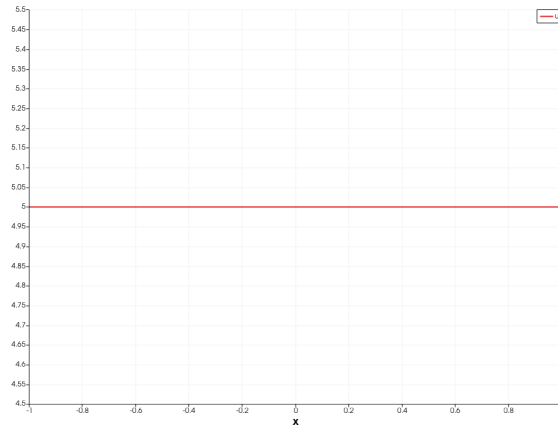
satisfying the following:

$$\frac{\partial u}{\partial t} - \nabla u = 0.$$

Boundary data and the initial condition are chosen consistently so that $u_{\text{exact}} = 5$ on every edge of Γ . This test checks the basic assembly procedure and control flow of the code for a trivial case. At final time $T = 1$, the numerical solution remains constant on the whole graph, with value $u = 5$ up to roundoff errors. In this simple case, the numerical algorithms are expected to return the exact solution. This is indeed the case, as shown in the following Figure. A mesh with 100 elements per edge is used to discretize the graph. The graph is discretized again using 100 elements per edge. The final time of observation is $T = 1$, the time step is $\Delta t = 0.05$ and the Backward Euler scheme is used.



(a) Solution on the star graph.



(b) Restriction of the numerical solution along the horizontal path $x \in (-1, 1)$, corresponding to the union of the two edges $\{v_0, v_1\}$ and $\{v_0, v_3\}$.

Figure 1: Test E1. Constant solution at final time $T = 1$. The solution is preserved on the whole graph and the one-dimensional restriction confirms that $u = 5$ along the selected branch.

3.1.2 Test E2: linear function

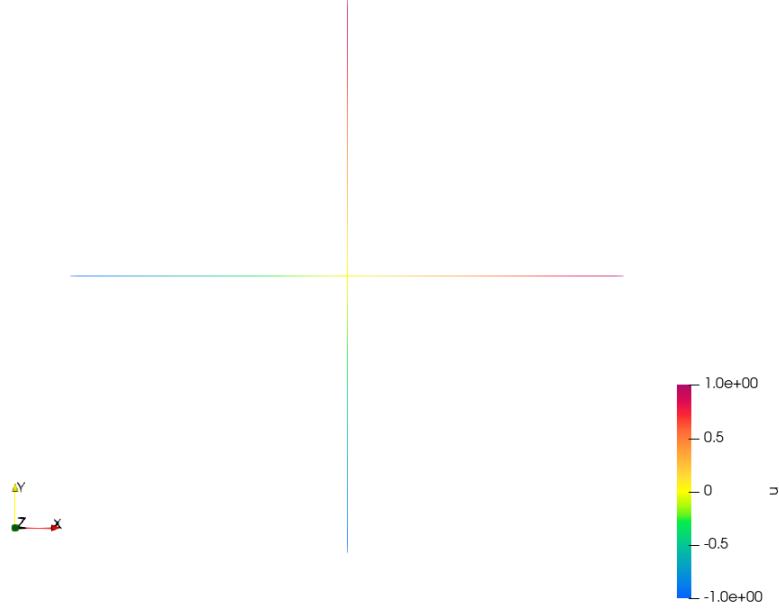
Let us consider the parabolic problem

$$\begin{cases} \frac{\partial u}{\partial t} - \Delta u = 0 & \text{in } \Gamma \times (0, T), \\ \sum_{e \in \mathcal{E}_{v_0}} \frac{\partial u_e}{\partial n_e}(v_0) = 0, \\ u(v_1, t) = 1, & t \in (0, T), \\ u(v_2, t) = 1, & t \in (0, T), \\ u(v_3, t) = -1, & t \in (0, T), \\ u(v_4, t) = -1, & t \in (0, T), \\ u(x, y, 0) = x + y, & \text{on } \Gamma. \end{cases}$$

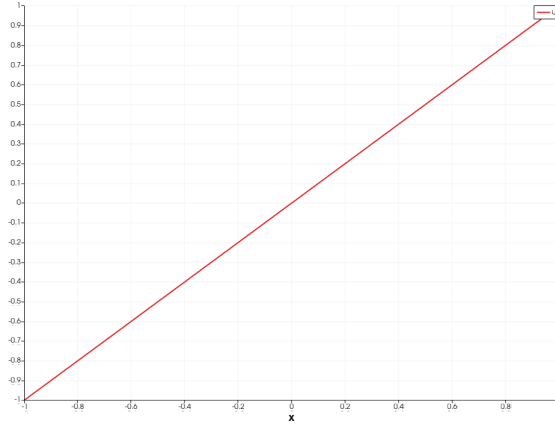
The exact solution is stationary and is given by

$$u_{\text{ex}}(x, y) = x + y.$$

Since the exact solution is linear on each edge, it belongs to the finite element space of continuous piecewise linear functions. Therefore, the numerical method is expected to reproduce it up to roundoff errors. The graph is discretized again using 100 elements per edge. The final time is still $T = 1$, the time step is $\Delta t = 0.05$ and the Backward Euler scheme is used. This benchmark proves that our code can handle various boundary conditions.



(a) Solution on the star graph.



(b) Restriction of the numerical solution along the horizontal path $x \in (-1, 1)$, corresponding to the union of the two edges $\{v_0, v_1\}$ and $\{v_0, v_3\}$.

Figure 2: Test E2. Linear solution at final time $T = 1$. The solution is preserved on the whole graph and the one-dimensional restriction confirms that $u = x$ along the selected branch.

3.1.3 Test E3: function of a single variable

The third elliptic test considers a non-polynomial function depending only on the horizontal coordinate:

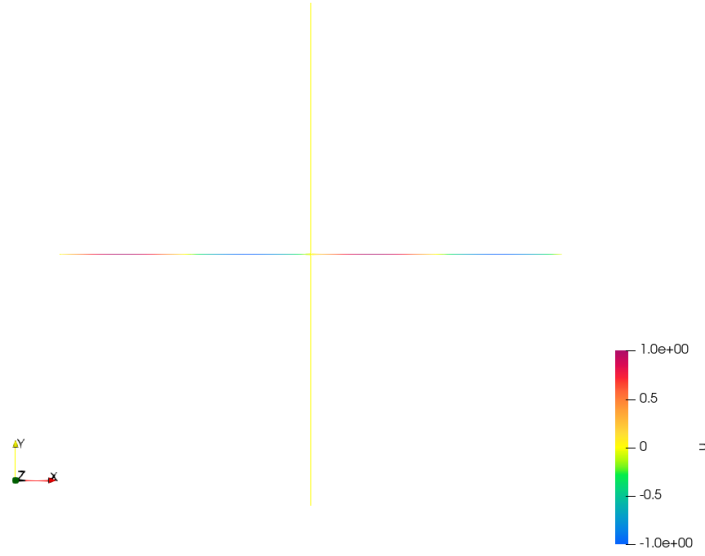
$$u_{\text{ex}}(x, y) = \sin(2\pi x).$$

This solution is not exactly represented by the piecewise linear finite element space. Therefore, unlike the previous two tests, the numerical error is nonzero and should decrease as the mesh is refined. This test is useful to observe the spatial convergence properties of the method.

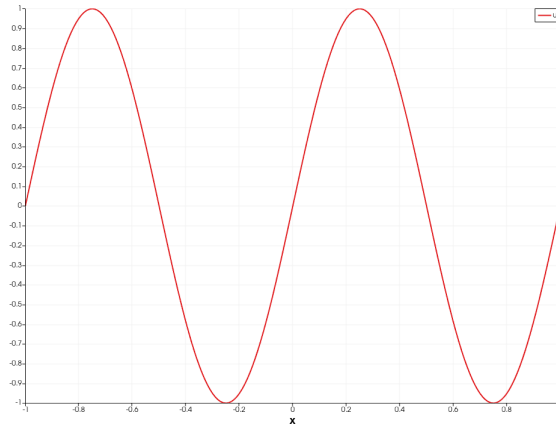
On the star-shaped graph Γ , we solve

$$\begin{cases} \partial_t u - \Delta u = 4\pi^2 \sin(2\pi x) & \text{in } \Gamma \times (0, T), \\ u(v, t) = 0 & \forall v \in \mathcal{V}, t \in (0, T), \\ u(x, y, 0) = \sin(2\pi x) & \text{on } \Gamma. \end{cases}$$

Unlike the constant and linear benchmarks, the exact solution is not contained in the finite element space of continuous piecewise linear functions. Therefore the numerical error is not expected to vanish, but it should decrease as the mesh is refined. The final time is $T = 1$, the time step is $\Delta t = 0.005$ and the Backward Euler method is used.



(a) Numerical solution on the star graph. Notice that the discrete solution is correctly globally continuous, constant along the y direction with value $u(0, y) = 0$



(b) Restriction of the solution along the horizontal path $x \in (-1, 1)$.

Figure 3: Test E3. Numerical solution associated with $u_{\text{ex}}(x, y, t) = \sin(2\pi x)$. The solution is zero on the vertical branches, where $x = 0$, and is globally continuous at the central vertex.

To study the spatial convergence, the problem is solved using 10, 20, 40, and 80 elements per edge. The time step is chosen as $\Delta t = 0.025h^2$, so that the temporal error is negligible with respect to the spatial error. Since continuous piecewise linear finite elements are used, the interpolation error for a sufficiently regular exact solution satisfies the classical estimate: $\|u - u_h\|_{L^2(\Gamma)} \leq Ch^2\|u\|_{H^2(\Gamma)}$. Therefore, when the temporal error is made negligible by choosing $\Delta t = 0.025h^2$, the dominant contribution to the error is the spatial finite element error. Hence, second-order convergence in the L^2 -norm is expected. The numerical results confirm this behaviour, since the observed convergence rates are approximately equal to 2.

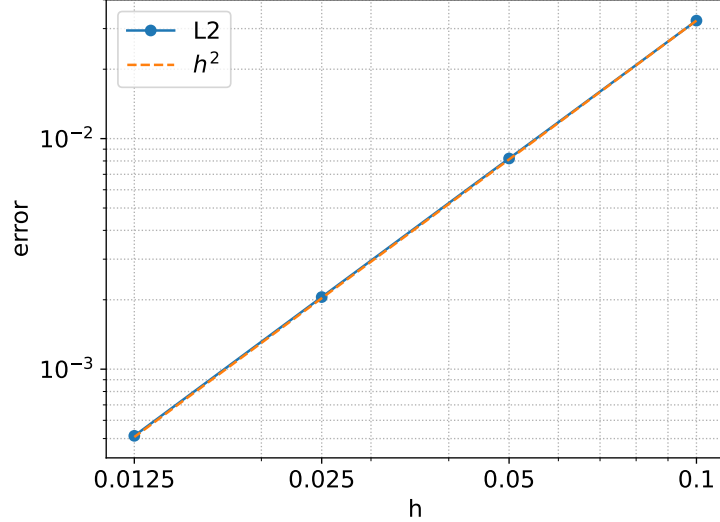


Figure 4: Test E3. Spatial convergence for the parabolic sine benchmark. The results show second-order convergence in the L^2 -norm.

3.2 Parabolic Test Cases

We now present the parabolic benchmarks implemented in this project.

3.2.1 Test 1:

This benchmark introduces a genuinely time-dependent exact solution. We solve

$$\partial_t u - \Delta u + u = 0 \quad \text{on } \Gamma \times (0, T).$$

The exact solution on each edge is

$$u_{\text{ex}}(s, t) = \exp \left[- \left(\frac{\pi^2}{4} + 1 \right) t \right] \cos \left(\frac{\pi s}{2} \right).$$

Homogeneous Dirichlet conditions are imposed at the four external vertices, while the central vertex satisfies the natural Kirchhoff condition. This benchmark validates the time-dependent part of the solver.

For the temporal convergence test, a fine spatial mesh with 320 elements per edge is used. Since each edge has unit length, the mesh size is $h = \frac{1}{320} = 0.003125$. This

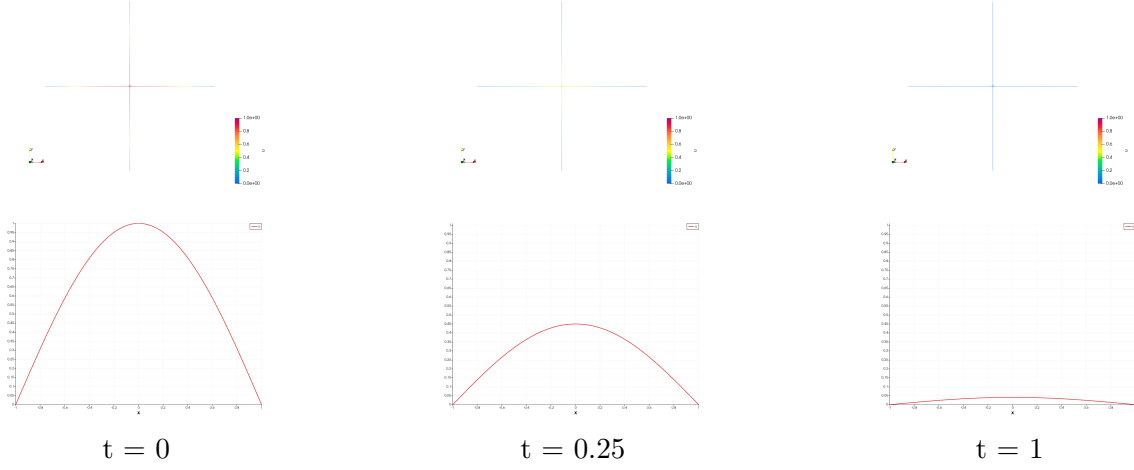


Figure 6: This test considers a radially decaying solution where both the diffusion and reaction term contribute to the temporal dynamics of the solution. The Backward Euler scheme is used and a mesh size $h = 0.01$.

choice makes the spatial error negligible with respect to the temporal error. Indeed, for piecewise linear finite elements the L^2 -error behaves as $O(h^2)$, and $h^2 \simeq 9.77 \cdot 10^{-6}$. On the other hand, the smallest time step used in the test is $\Delta t = 0.025$, for which $\Delta t^2 = 6.25 \cdot 10^{-4}$. Therefore, even for the second-order Crank–Nicolson method, the temporal error is expected to dominate the spatial one.

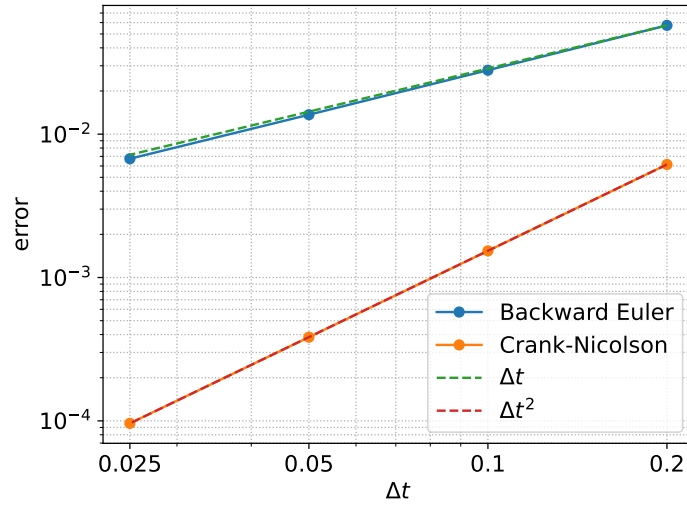


Figure 7: Test 4. Temporal convergence of Backward Euler and Crank–Nicolson. The results show first-order convergence for Backward Euler and second-order convergence for Crank–Nicolson.

3.3 Elliptic eigenvalue problems

The following numerical experiments consider the eigenvalue problem for the Laplacian operator on metric graphs, endowed with Neumann–Kirchhoff conditions at the vertices. Three different graph topologies are analyzed in order to compare our numerical results with the reference ones reported in [2]. The comparison is understood up to the sign and normalization of the eigenfunctions. Moreover, in the case of multiple eigenvalues, the computed eigenfunctions may differ by linear combinations within the same eigenspace.

3.3.1 Test 1: four-pointed star graph

The goal of this benchmark, as for the following ones, is to verify that the code correctly solves generalized eigenvalue problems on metric graphs. This also represents the first step of the parabolic pipeline, since an eigenfunction of the graph Laplacian provides an exact modal solution of the heat equation. The first topology considered is the four-pointed star graph. We compute the eigenpairs (λ, u) satisfying

$$-\Delta u = \lambda u \quad \text{on } \Gamma,$$

with homogeneous Neumann conditions at the four external vertices and the Neumann–Kirchhoff condition at the central vertex. Each edge has length 1 and is discretized using 100 finite elements. Fig. 8 shows the first six eigenmodes. The blue lines represent the metric graph, while the red curves represent the corresponding eigenfunctions lifted in the vertical direction. The eigenfunctions are rescaled only for visualization purposes; the eigenvalues are those computed by the finite element generalized eigenvalue problem. The results are in good agreement with the reference results reported on page 31 of [2].

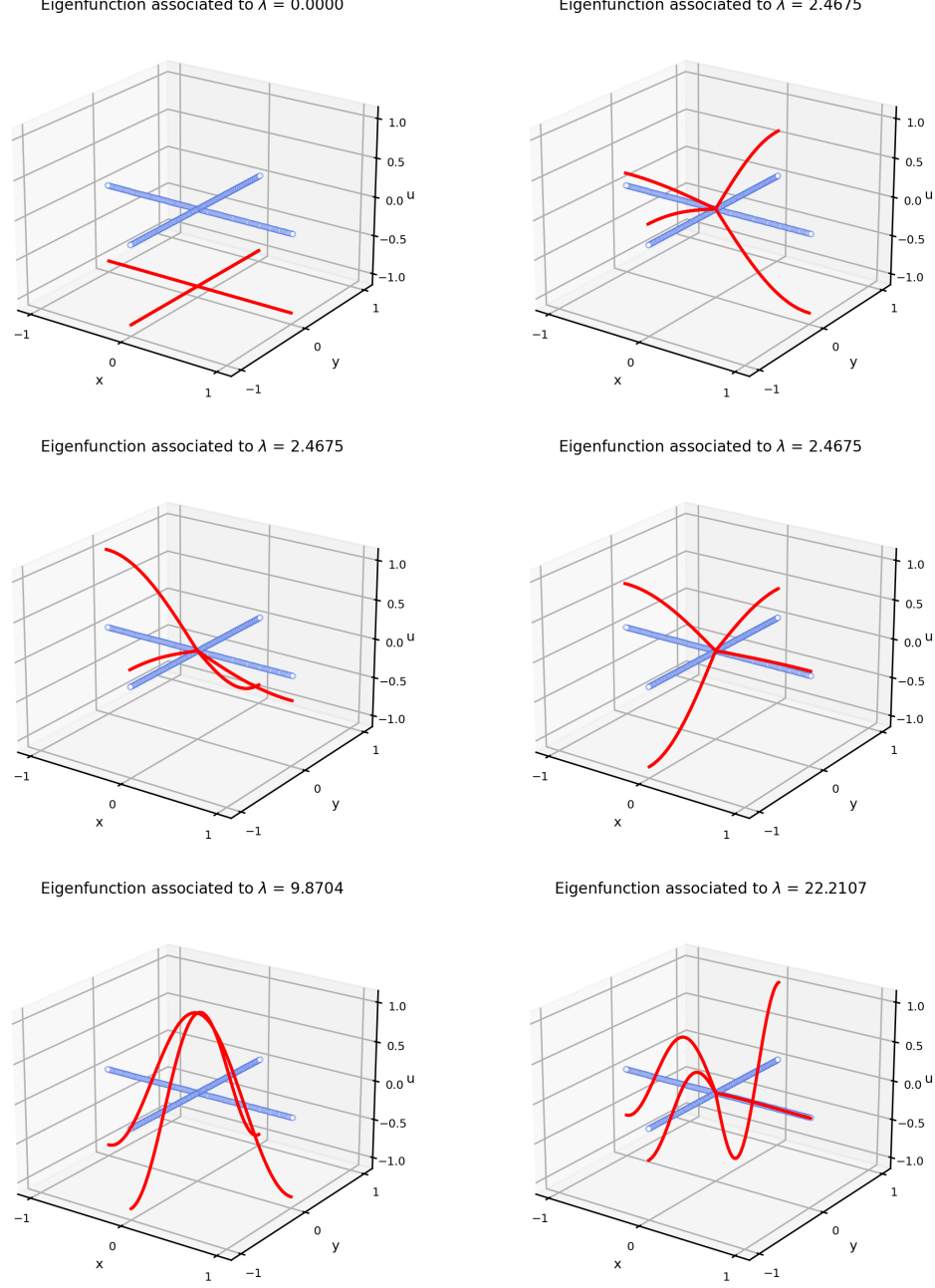


Figure 8: First six eigenmodes of the Laplacian on the four-pointed star graph. The blue graph represents the computational domain, while the red curves show the corresponding eigenfunctions. The vertical scaling is used only for visualization.

The eigenvectors computed by the finite-element solver are normalized with respect to the mass matrix, namely $\mathbf{u}^T M \mathbf{u} = 1$, which corresponds to the discrete $L^2(\Gamma)$ -normalization. In the figures, however, each eigenfunction is additionally rescaled by its maximum absolute nodal value only for visualization purposes.

3.3.2 Test 2: graphene-like graph

The second benchmark considers a graphene-like metric graph. This topology is more complex than the four-pointed star graph, since it contains several internal junctions and cycles. The aim is to test our solver on a graph with a richer

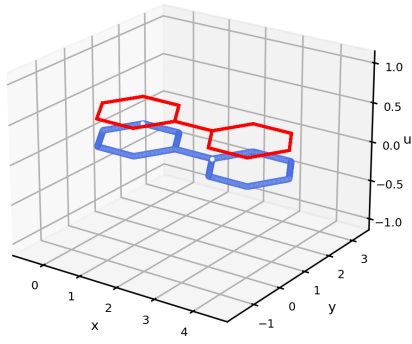
connectivity structure, while keeping the same differential operator and vertex conditions introduced in the previous test. We compute the eigenpairs (λ, u) satisfying

$$-\Delta u = \lambda u \quad \text{on } \Gamma,$$

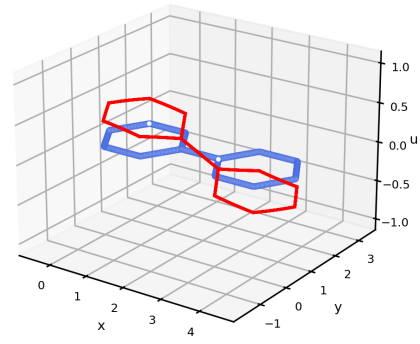
with homogeneous Neumann conditions at the external vertices and Neumann–Kirchhoff conditions at the internal vertices. The graph has 12 vertices and 13 edges. Each edge has length 1 and is discretized using 100 elements 100 per edge.

Fig. 9 shows the first six eigenmodes, while Figure 10 shows the following four. The blue lines still represent the metric graph, whereas the red curves represent the computed eigenfunctions lifted in the vertical direction. As in the previous test, the vertical scaling is used only for visualization purposes. The two related images show good agreement with [2].

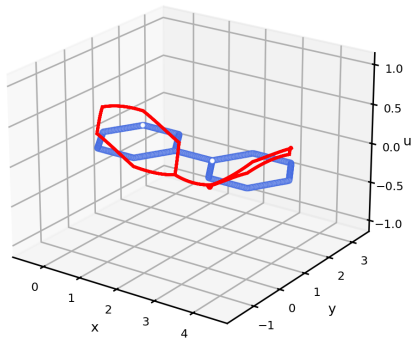
Eigenfunction associated to $\lambda = 0.0000$



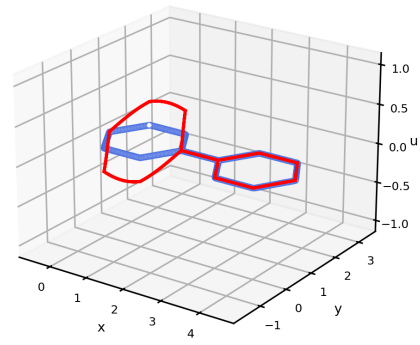
Eigenfunction associated to $\lambda = 0.1571$



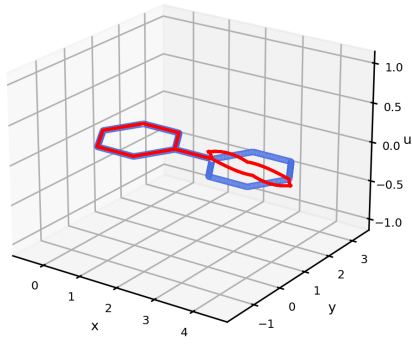
Eigenfunction associated to $\lambda = 0.9257$



Eigenfunction associated to $\lambda = 1.0966$



Eigenfunction associated to $\lambda = 1.0966$



Eigenfunction associated to $\lambda = 1.5608$

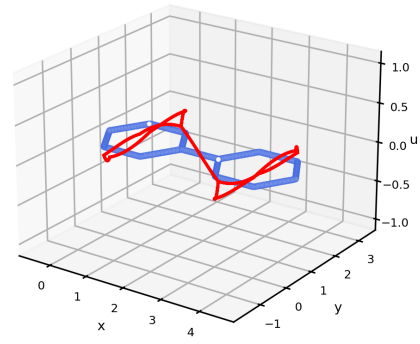


Figure 9

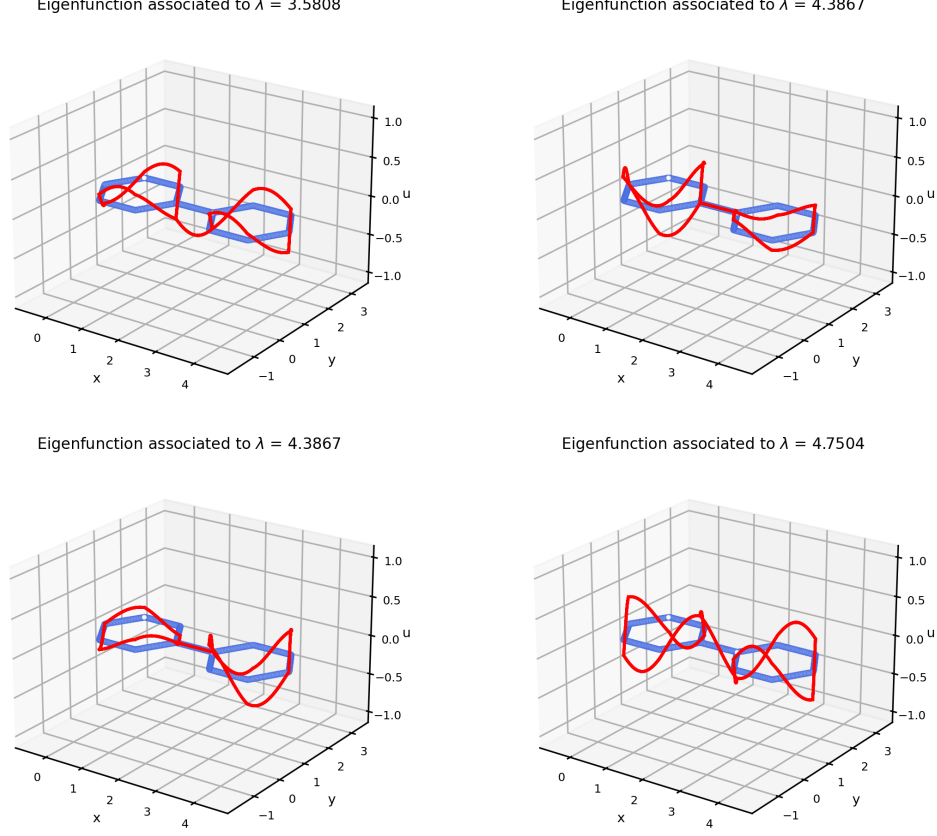


Figure 10: Fig. 9 and 10 show the first ten eigencouples of the Laplacian operator on a graphene-like graph. In blue, the domain of the problem. Results are scaled.

3.3.3 Test 3: binary tree graph

The third benchmark considers a binary tree metric graph. This topology is particularly relevant because it contains several branching points and many terminal vertices, while remaining simpler than a general network. It is therefore a useful intermediate test between elementary graph topologies and more complex network-like domains. We compute again the eigenpairs (λ, u) satisfying

$$-\Delta_{\Gamma} u = \lambda u \quad \text{on } \Gamma,$$

with homogeneous Neumann conditions at the leaves and Neumann–Kirchhoff conditions at the internal ones. The graph has 16 vertices and 15 edges. Each edge has length 1 and is discretized using 100 finite elements as in the previous cases.

Fig. 11 and Fig. 12 show the first sixteen eigenmodes. As before, the blue lines represent the metric graph and the red curves represent the eigenfunctions lifted in the vertical direction. The vertical scaling is used only for visualization. This experiment shows a very good agreement with the reference spectral behaviour reported in [2].

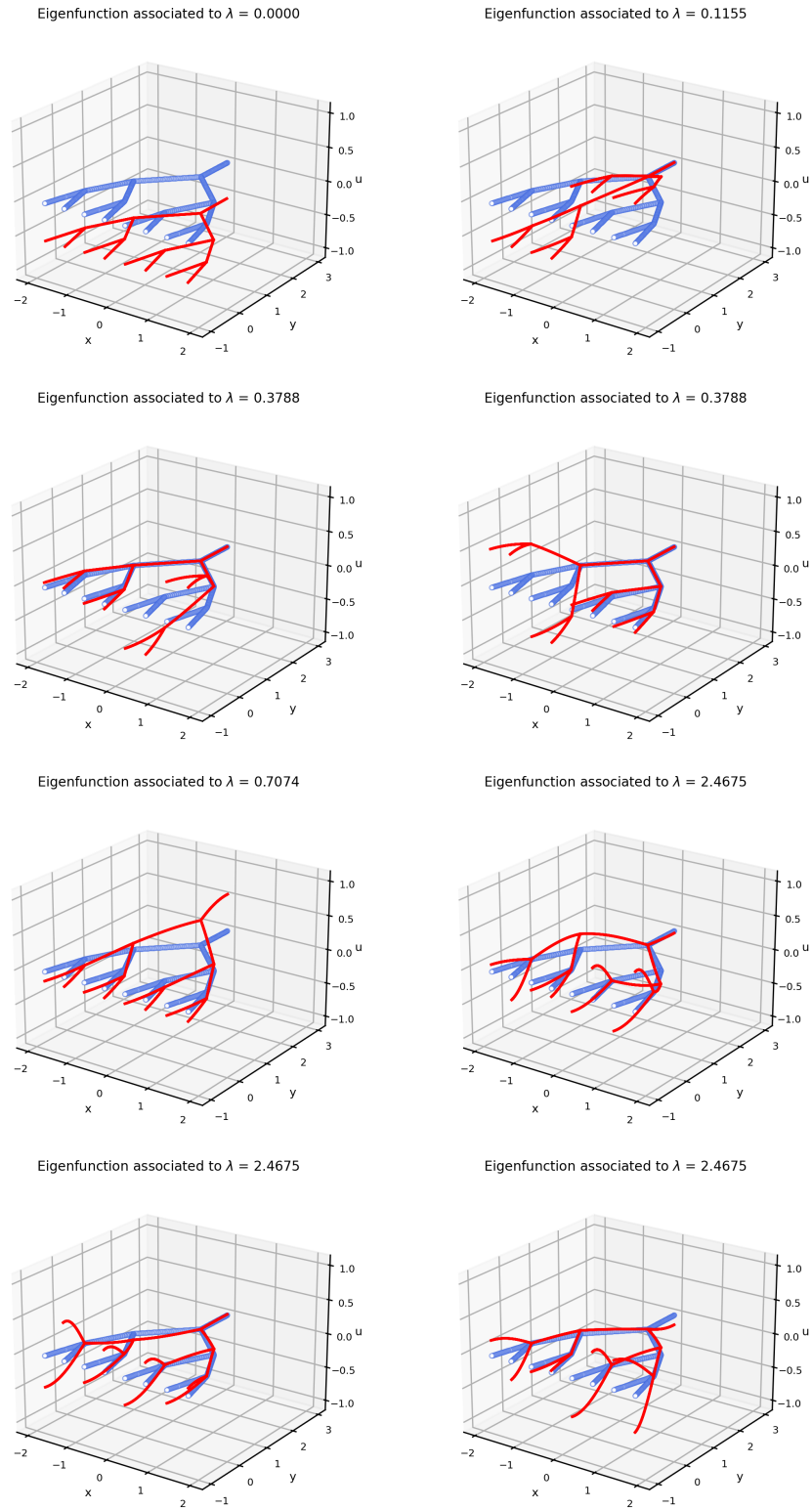


Figure 11: First eight eigenmodes of the Laplacian on the binary tree graph.

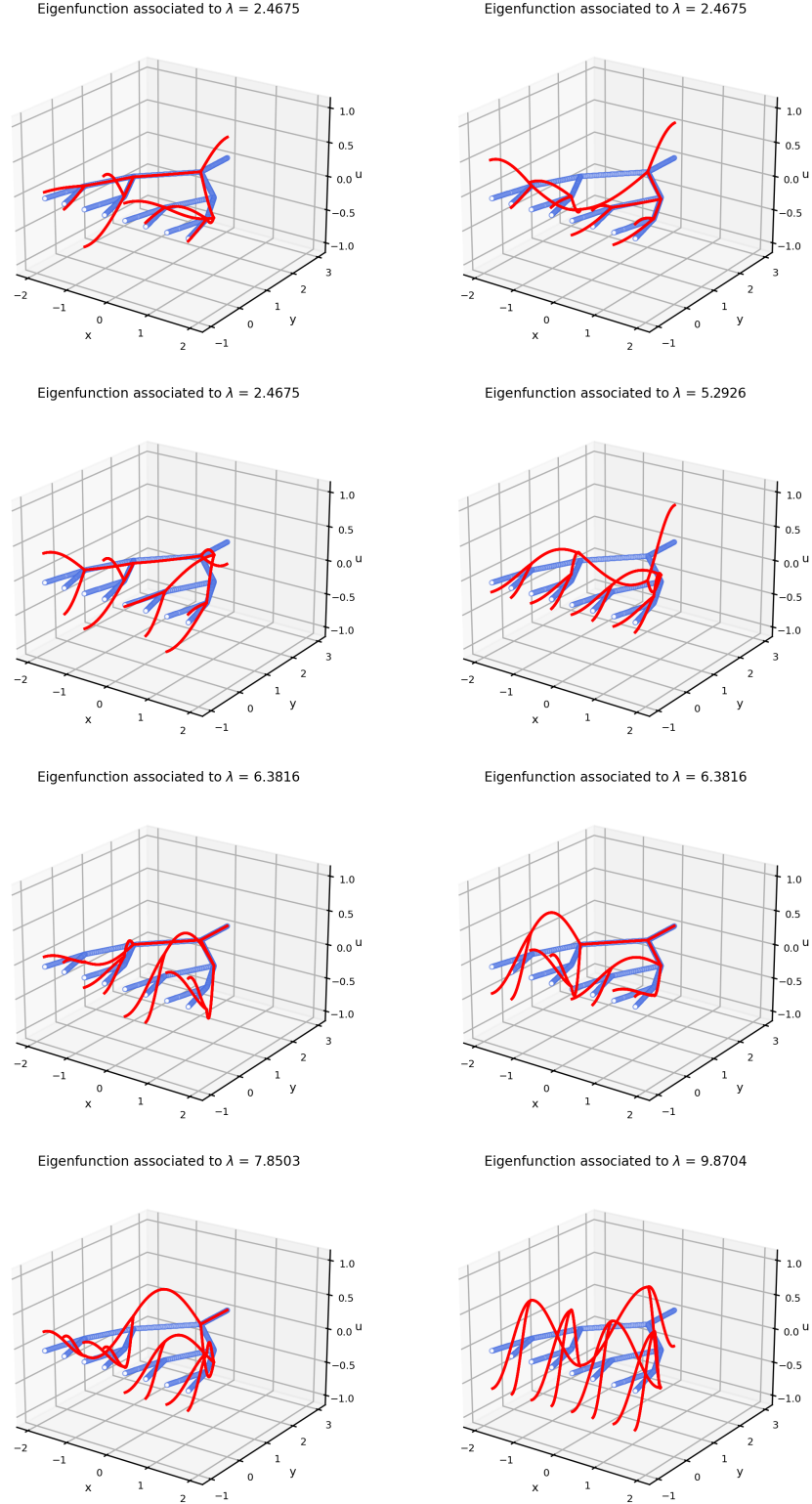


Figure 12: Eigenmodes 8–15 of the Laplacian on the binary tree graph.

Remark. In all the experiments, eigenfunctions associated with larger eigenvalues display a progressively more oscillatory behaviour along the graphs. This is consistent with the spectral properties of the Laplacian operator: the operator is unbounded, meaning that its spectrum is not bounded from above and $\lambda_k \rightarrow +\infty$ as $k \rightarrow \infty$;

therefore, higher eigenvalues correspond to higher-frequency modes on the metric graph.

3.4 Parabolic eigenvalue problems

The eigenvalue computations of the previous section can be used to construct exact solutions for the heat equation on metric graphs. Let u_k be an eigenmode of the graph Laplacian associated with the eigenvalue λ_k , namely

$$-\Delta u_k = \lambda_k u_k \quad \text{on } \Gamma,$$

with Neumann–Kirchhoff vertex conditions. Then the heat equation

$$\partial_t u - \mu \Delta u = 0 \quad \text{on } \Gamma \times (0, T]$$

admits the exact modal solution

$$u(x, t) = e^{-\mu \lambda_k t} u_k(x).$$

This provides a useful validation test for the parabolic solver, since both the spatial structure and the temporal decay are known explicitly. The initial datum is a normalized eigenmode, so that

$$\|u_k\|_{L^2(\Gamma)} = 1.$$

Consequently, the exact evolution of the L^2 -norm is

$$\|u(t)\|_{L^2(\Gamma)} = e^{-\mu \lambda_k t}.$$

In the following tests we take $\mu = 1.0$, $T = 1$, $\Delta t = 0.01$, and use the Crank–Nicolson method.

3.4.1 Test 4: heat equation on the graphene-like graph

For the graphene-like graph we select the sixth eigenmode, associated with $\lambda_6 \simeq 3.5808$. Fig. 13 shows the numerical solution at selected time instants. The shape of the selected eigenmode is preserved, while its amplitude decays in time. Fig. 14 compares the numerical L^2 -norm with the exact decay law. Since the initial eigenmode is mass-normalized, the curve starts from 1. At $T = 1.0$, the numerical norm is approximately $2.7842 \cdot 10^{-2}$, in excellent agreement with the exact value $2.7853 \cdot 10^{-2}$.

3.4.2 Test 5: heat equation on the binary tree graph

For the binary tree graph we select the fifth eigenmode, associated with $\lambda_5 \simeq 2.4675$. Fig. 15 shows the time evolution of the numerical solution. Also in this case, the eigenmode profile is preserved and only its amplitude decays. Fig. 16 shows the agreement between the numerical L^2 -norm and the exact exponential decay. At $T = 1.0$, the numerical norm is approximately $8.4790 \cdot 10^{-2}$, while the exact value is $8.4801 \cdot 10^{-2}$.

Heat equation on graphene, sixth eigenmode, $\lambda = 3.5808$

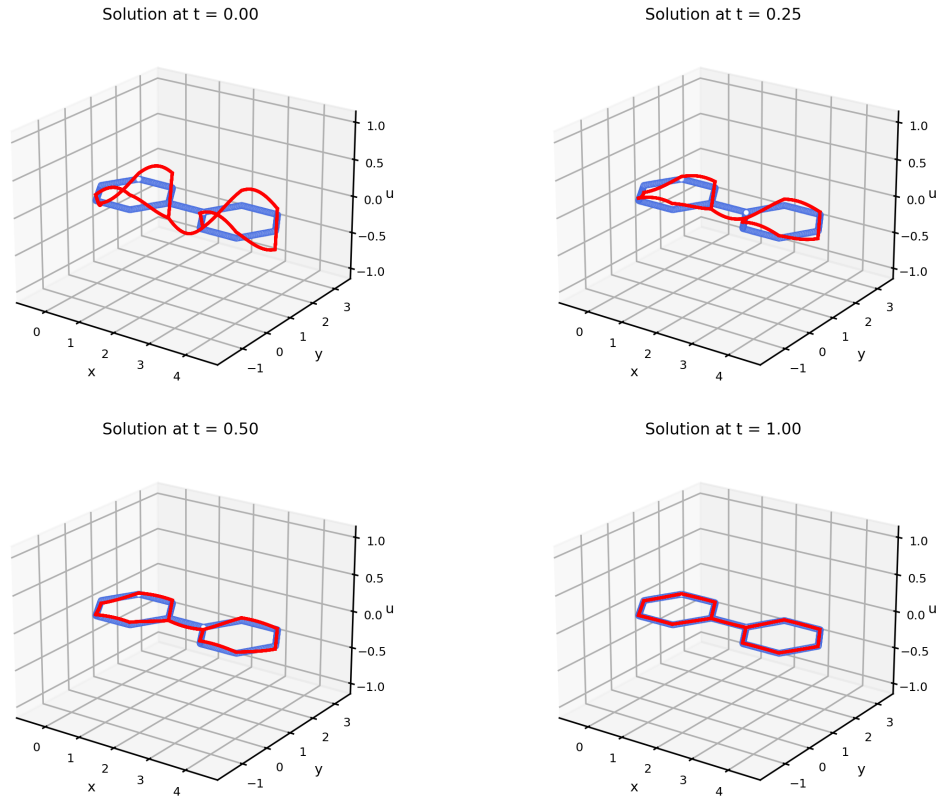


Figure 13: Heat equation on the graphene-like graph with initial datum given by the sixth eigenmode, associated with $\lambda_6 \simeq 3.5808$.

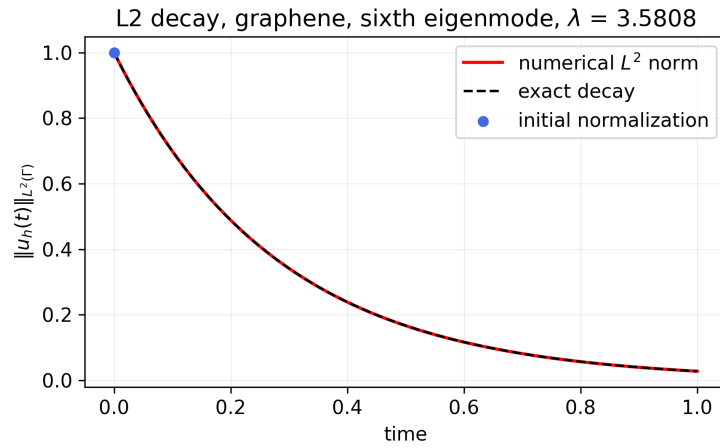


Figure 14: Decay of the numerical L^2 -norm compared with the exact law for the graphene-like graph.

Heat equation on tree, fifth eigenmode, $\lambda = 2.4675$

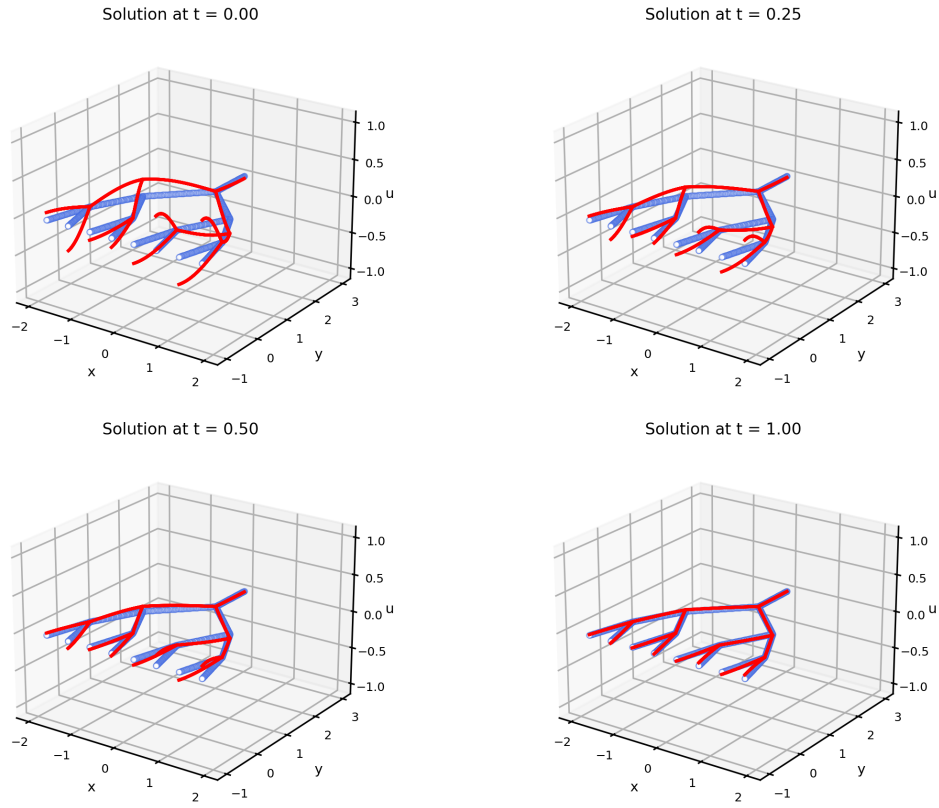


Figure 15: Heat equation on the binary tree graph with initial datum given by the fifth eigenmode, associated with $\lambda_5 \simeq 2.4675$.

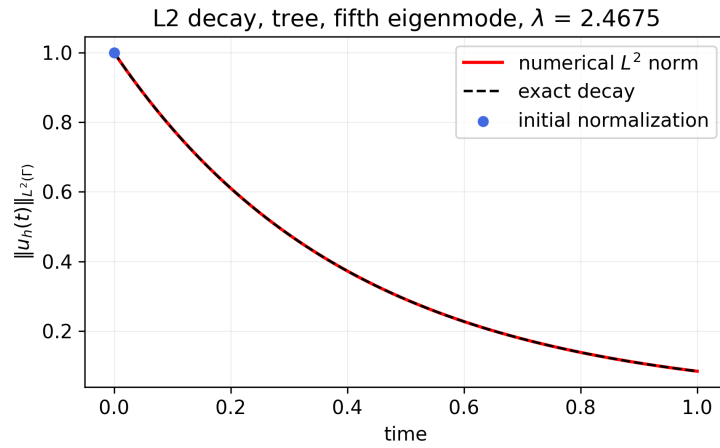


Figure 16: Decay of the numerical L^2 -norm compared with the exact law for the binary tree graph.

3.4.3 Laplace matrix spectral comparison

It can be interesting to compare the spectrum of the combinatorial Laplacian of the original graph Γ with the spectrum of the Laplace matrix associated with the corresponding extended graph $\mathcal{G}$.

Given a graph with N vertices, the combinatorial Laplacian, introduced in Definition 1.6 (and described in App. A), is

$$L = D - A,$$

where D is the degree matrix and A is the adjacency matrix. In our implementation, this matrix is assembled through a custom C++ routine from the graph connectivity. Since L is real and symmetric, its eigenvalues are then computed using Eigen's `SelfAdjointEigenSolver()`.

For the extended graph, the finite element discretization of the Laplace eigenvalue problem leads to the generalized algebraic problem

$$H\mathbf{u} = \lambda M\mathbf{u},$$

where H is the stiffness matrix and M is the mass matrix. This problem is solved using Eigen's `GeneralizedSelfAdjointEigenSolver()`. This choice is motivated by the fact that H and M are symmetric matrices and M is positive definite.

The comparison is performed on three graph topologies: the four-pointed star graph, the graphene-like graph and a family of tree graphs. The extended graph is constructed using 100 finite elements per edge. This choice makes the spectral comparison consistent with the eigenfunction plots presented in the previous benchmarks.

Fig. 17 reports the comparison between the eigenvalues of the combinatorial Laplace matrix and those of the extended Laplace matrix. The corresponding numerical values are given in Tables 1, 2 and 3.

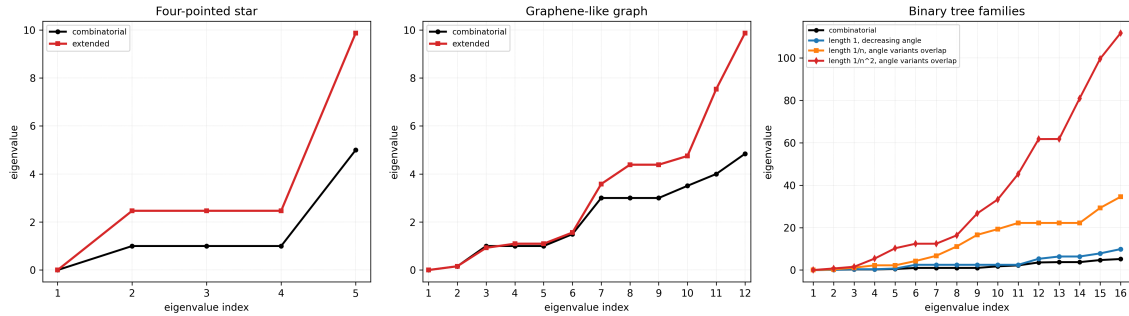


Figure 17: Comparison between the spectrum of the combinatorial Laplace matrix of the original graph and the spectrum of the extended Laplace matrix. Eigenvalues are plotted against their index. The presence of symmetries in the graph topology causes the existence of multiple eigenvalues.

For the tree graph, several geometric configurations are considered. The first case has constant edge length and decreasing bifurcation angle. The other cases have either linearly or quadratically (with the bifurcation level) decreasing edge lengths. More precisely, if n denotes the bifurcation level, the edge lengths are chosen as

$$\ell_n = 1, \quad \ell_n = \frac{1}{n}, \quad \ell_n = \frac{1}{n^2}.$$

The numerical experiments show that, for fixed and decreasing edge-length laws, changing the bifurcation angle does not affect the spectrum of the extended Laplace

Combinatorial Laplace matrix	Extended Laplace matrix
0	0
1.0000	2.4675
1.0000	2.4675
1.0000	2.4675
5.0000	9.8704

Table 1: Spectral comparison for the four-pointed star graph.

Combinatorial Laplace matrix	Extended Laplace matrix
0	0
0.1578	0.1571
1.0000	0.9257
1.0000	1.0966
1.0000	1.0966
1.4931	1.5608
3.0000	3.5808
3.0000	4.3867
3.0000	4.3867
3.5069	4.7504
4.0000	7.5371
4.8422	9.8704

Table 2: Spectral comparison for the graphene-like graph.

matrix. This is expected, since the metric graph Laplacian depends on the graph topology and on the edge lengths, while the planar drawing angles only affect the visualization of the graph.

The results show that the spectra of the combinatorial and extended Laplace matrices are generally different in value. Nevertheless, the comparison is useful to observe how the graph topology influences the low-frequency part of the spectrum. In the tree cases, the dominant effect is produced by the edge lengths: decreasing edge lengths lead to larger eigenvalues, while changing the bifurcation angles does not modify the spectrum.

An additional observation concerns the effect of decreasing edge lengths. When the tree contains shorter and shorter edges, the eigenvalues of the extended Laplace matrix increase significantly. This is consistent with the metric nature of the operator: small edges allow eigenfunctions with rapid spatial oscillations and these high-frequency modes correspond to larger eigenvalues. Therefore, the growth of the spectrum reflects the increasing presence of small geometric scales in the graph.

Comb.	$\ell_n = 1$	$\alpha = \pi/4, \ell_n = 1/n$	$\alpha = \pi/4, \ell_n = 1/n^2$
0	0	0	0
0.0968	0.1155	0.3303	0.7426
0.2679	0.3788	1.1186	1.5485
0.2679	0.3788	2.2205	5.4138
0.4965	0.7074	2.2205	10.2825
1.0000	2.4675	4.2573	12.4108
1.0000	2.4675	6.7012	12.4108
1.0000	2.4675	11.1094	16.3523
1.0000	2.4675	16.5789	26.6628
1.7356	2.4675	19.2819	33.2463
2.1939	2.4675	22.2071	45.2803
3.5767	5.2926	22.2071	61.7503
3.7321	6.3816	22.2071	61.7907
3.7321	6.3816	22.2071	80.9487
4.7093	7.8503	29.2923	99.6367
5.1912	9.8704	34.5652	111.7665

Table 3: Spectral comparison for the tree graph. The omitted angle variants give the same extended spectra for fixed edge-length laws.

4 Prion-like spreading of neurodegenerative diseases within the brain’s connectome

We have validated our finite element library for the solution of graph-based differential equations on a sequence of elliptic and parabolic benchmark problems. We now consider a more application-oriented test case. In particular, we study the Fisher–Kolmogorov equation on a brain connectome’s topology. This problem provides a natural extension of the previous linear parabolic models, as it combines diffusion along the graph with a nonlinear reaction term. At the same time, it provides a meaningful benchmark for assessing the scalability of our framework through graph reordering and parallel programming techniques, with the particularly interesting perspective of modeling the propagation of infectious misfolded proteins through the brain, a process that is responsible for neurodegenerative diseases. The goal of this section is therefore not to provide a detailed biological study, but to investigate how the developed solver can be extended from controlled benchmark problems to a realistic network-based diffusion-reaction model.

4.0.1 One-dimensional sensitivity analysis

Prion diseases are characterized by a chain reaction in which misfolded proteins force native ones into a similar pathogenic structure. A physics-based reaction-diffusion model can explain the growth and spreading of misfolded protein in neurodegenerative diseases, like Alzheimer’s disease. To characterize the progression of misfolded proteins across the brain, we combine the classical Fisher–Kolmogorov equation for population dynamics with an anisotropic diffusion model. Following the work described in [7], we perform a systematic sensitivity analysis, demonstrating the role of the individual model parameters. We show how the misfolded protein concentration is sensitive to the coefficients of growth and extracellular diffusion. We solve

$$\frac{\partial c}{\partial t} - d \frac{\partial^2 c}{\partial x^2} = \alpha c(1 - c), \quad (x, t) \in (-1, 1) \times (0, 20],$$

with homogeneous Neumann boundary conditions,

$$\frac{\partial c}{\partial x}(-1, t) = \frac{\partial c}{\partial x}(1, t) = 0.$$

The initial concentration is zero at every mesh node except at the central node, where $c(0,0) = 0.1$. The interval is discretized using 200 linear finite elements. Backward Euler is employed with $\Delta t = 0.1$, and the nonlinear problem arising at each time step is solved by Newton's method.

The sensitivity analysis combines the diffusion coefficients

$$d \in \{10^{-4}, 2 \cdot 10^{-4}, 4 \cdot 10^{-4}\}$$

with the growth coefficients

$$\alpha \in \{1, 2, 4\}.$$

Figure 18 presents the results using an activation-time representation. For every pair (x, c) , the color denotes the first time at which the concentration at position x reaches the prescribed level c . The black curves are therefore contours of equal activation time.

Increasing α accelerates the local conversion of healthy proteins, whereas increasing d enhances their spatial propagation. Consequently, both parameters increase the propagation speed, although through different mechanisms. The numerical fronts remain symmetric with respect to $x = 0$, and their behavior is consistent with the Fisher–KPP asymptotic relation

$$v_{\text{front}} = 2\sqrt{d\alpha}.$$

The experiment therefore verifies that the finite element solver correctly captures the combined nonlinear growth and diffusion mechanisms before its application to a brain connectome.



images/fisher_kolmogorov_1d_sensitivity.png

Figure 18: Activation-time maps for the one-dimensional Fisher–Kolmogorov problem. The growth coefficient increases from left to right, while the diffusion coefficient increases from top to bottom. For each pair (x, c) , the color represents the first time at which the numerical concentration at x reaches the level c ; black curves denote equal activation times. All panels use the same time scale $t \in [0, 20]$.

5 C++ implementation

6 Conclusions

A Properties of the Combinatorial Laplacian

The combinatorial Laplacian $L_{\mathcal{G}} = MM^{\top}$ satisfies the following properties:

- Symmetry

$$L_{\mathcal{G}}^{\top} = (MM^{\top})^{\top} = (M^{\top})^{\top}M^{\top} = MM^{\top} = L_{\mathcal{G}}.$$

- Positive semidefiniteness

$$x^{\top}L_{\mathcal{G}}x = x^{\top}MM^{\top}x = (M^{\top}x)^{\top}(M^{\top}x) = \|M^{\top}x\|_2^2 \geq 0,$$

hence $x^{\top}L_{\mathcal{G}}x \geq 0$ for all $x \in \mathbb{R}^V$.

- Singularity

$$(M^{\top}f)_e = f(v_i) - f(v_j) \quad \forall e = \{v_i, v_j\} \in \mathcal{E}, f \in \mathbb{R}^V$$

So $(M^{\top}f)_e = 0$ if and only if $f(v_i) = f(v_j)$ that is, f must be constant on every pair of adjacent vertices. If $\mathcal{G}$ is connected, this forces f to be globally constant, so

$$\ker(M^{\top}) = \text{span}\{(1, 1, \dots, 1)^{\top}\}.$$

Since $L_{\mathcal{G}} = MM^{\top}$, we have $L_{\mathcal{G}}f = 0 \iff M^{\top}f = 0$, hence

$$\ker(L_{\mathcal{G}}) = \ker(M^{\top}) = \text{span}\{(1, 1, \dots, 1)^{\top}\},$$

confirming that $L_{\mathcal{G}}$ is singular. More generally, the constant function argument applies independently on each connected component: a vector f satisfying $M^{\top}f = 0$ must be constant within each component but may take different values across components. Therefore

$$\dim \ker(L_{\mathcal{G}}) = k,$$

where k is the number of connected components of $\mathcal{G}$ ¹.

- Meaning

$$(Mg)_i = \sum_{e \ni v_i} \pm g_e, \forall g \in \mathbb{R}^E$$

where the sign is +1 if the flux along e is outgoing from v_i and -1 if it is incoming to v_i , reflecting the local flow configuration at its incident edges. This is the discrete analogue of a divergence: M accumulates the signed flux of g at each node.

Dually, for $f \in \mathbb{R}^V$, the action of $M^{\top}$ on edge $e = \{v_i, v_j\}$ reads

$$(M^{\top}f)_e = f(v_i) - f(v_j),$$

which is the difference of the nodal values at the endpoints of e , i.e. the discrete analogue of a derivative. Thus $M^{\top}$ plays the role of a *discrete gradient*.

Since M is a discrete divergence and $M^{\top}$ a discrete gradient, their composition $L_{\mathcal{G}} = MM^{\top}$ is naturally interpreted as a *discrete Laplacian*, which motivates calling $L_{\mathcal{G}}$ the Laplacian of the combinatorial graph $\mathcal{G}$.

Input graphs are specified in a simple `.graph` plain-text format: the first line gives $|\mathcal{V}|$ and $|\mathcal{E}|$, followed by one line per edge with endpoint indices and length.

¹In a (possibly disconnected) graph, a connected component is defined as a maximal connected subgraph, i.e. a largest connected subset of vertices (together with the edges between them).

B Derivation of the weak and algebraic formulations

B.1 Problem formulation

A metric graph $\Gamma = (\mathcal{V}, \mathcal{E})$ consists of a finite set of vertices and a finite set of edges. The vertex set is generally partitioned into three disjoint subsets:

$$\mathcal{V} = \mathcal{T} \cup \mathcal{V}_1 \cup \mathcal{V}_2,$$

where $\mathcal{T}$ collects the vertices of degree one (i.e. they are the terminal points of the graph), $\mathcal{V}_1$ collects the *simple junction* vertices, at which the flux is conserved and $\mathcal{V}_2$ collects the *generalized junction* vertices at which a different local dynamics can be prescribed. On Γ we consider a system of parabolic problems. Each line of this system lives within an edge e and communicates with the others through boundary conditions.

For each edge $e \in \mathcal{E}$ and $t \in (0, T]$, the unknown $u_e(s, t) = u(s_e, t)$ satisfies

$$\frac{\partial u_e}{\partial t} - \frac{\partial}{\partial s} \left(\mu_e \frac{\partial u_e}{\partial s} \right) + r_e u_e = f_e, \quad 0 < s_e < \ell_e, \quad \forall e \in \mathcal{E}, \quad (9)$$

where $\mu_e = \mu_e(s, t) > 0$ is the diffusion coefficient, $r_e = r_e(s, t) \geq 0$ is the reaction coefficient and $f_e = f_e(s, t)$ is a given source term on e . The initial condition on each edge is the following

$$u_e(s, 0) = u_e^0(s), \quad 0 < s < \ell_e, \forall e \in \mathcal{E}.$$

At each terminal vertex, either a Dirichlet boundary condition or a homogeneous Neumann condition can be imposed. At each simple junction vertex $v_p \in \mathcal{V}_1$, the solution is continuous and the total flux is conserved:

$$\sum_{e \in \mathcal{E}_{v_p}} \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e} \right]_{v_p} = 0, \quad \forall v_p \in \mathcal{V}_1, \quad (10)$$

where $\mathcal{E}_{v_p}$ denotes the set of edges incident to v_p and $[\partial/\partial \mathbf{n}_e]_{v_p}$ denotes the outward normal derivative along e with respect to v_p . At each generalized junction vertex $v_q \in \mathcal{V}_2$, a local variable $u_q = u_q(t)$ is associated with the vertex and satisfies

$$c_q \frac{du_q}{dt} + r_q u_q = \sum_{e \in \mathcal{E}_{v_q}} \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e} \right]_{v_q} + f_q, \quad \forall v_q \in \mathcal{V}_2, \quad (11)$$

where $c_q > 0$ is a relaxation coefficient, $f_q = f_q(t)$ is a local source term, and $[\partial/\partial \mathbf{n}_e]_{v_q}$ is the outward normal derivative along each incident edge with respect to v_q . These are two different versions of the so-called Kirchhoff conditions.

B.2 Weak formulation

We consider the parabolic problem (9) on the metric graph $\Gamma = (\mathcal{V}, \mathcal{E})$, together with the Kirchhoff conditions (10)–(11), homogeneous Neumann boundary conditions at

the terminal vertices and the global continuity constraints. The spatial differential operator acting on each edge is

$$\mathcal{L}_e u_e = -\frac{\partial}{\partial s} \left(\mu_e \frac{\partial u_e}{\partial s} \right) + r_e u_e, \quad (12)$$

so that (9) reads $\partial_t u_e + \mathcal{L}_e u_e = f_e$ on each $e \in \mathcal{E}$. Note that when μ_e is constant and $r_e = V_e(s)$ is a potential, $\mathcal{L}_e$ reduces to the quantum-graph Hamiltonian $\mathcal{H}_e u_e = -\mu_e \frac{\partial^2 u_e}{\partial s^2} + v_e u_e$.

The weak formulation is obtained by multiplying (9) by a test function $w \in H^1(\Gamma)$, integrating over each edge, summing over $\mathcal{E}$ and integrating by parts the diffusion term. At each junction vertex, the boundary terms arising from integration by parts are replaced using the Kirchhoff conditions (10)–(11), which cancel for $v_p \in \mathcal{V}_1$ and contribute a local ODE term for $v_q \in \mathcal{V}_2$. Also contributions coming from terminal vertices cancel, as we consider homogeneous Neumann conditions. The weak problem reads:

Find $u(s, t) \in H^1(\Gamma)$ such that

$$\sum_{e \in \mathcal{E}} \int_e \left[\frac{\partial u_e}{\partial t} w_e \right] ds + a(u, w) - \sum_{v_q \in \mathcal{V}_2} c_q \frac{du_q}{dt} w_q = F(w) \quad \forall w \in H^1(\Gamma), \quad (13)$$

where the bilinear form $a : H^1(\Gamma) \times H^1(\Gamma) \rightarrow \mathbb{R}$ is given by

$$a(u, w) = \sum_{e \in \mathcal{E}} \int_e \left[\mu_e \frac{\partial u_e}{\partial s} \frac{\partial w_e}{\partial s} + r_e u_e w_e \right] ds - \sum_{v_q \in \mathcal{V}_2} r_q u_q w_q, \quad (14)$$

and the linear functional $F : H^1(\Gamma) \rightarrow \mathbb{R}$ is

$$F(w) = \sum_{e \in \mathcal{E}} \int_e f_e w_e ds - \sum_{v_q \in \mathcal{V}_2} f_q w_q. \quad (15)$$

B.3 Weak formulation

We start from the strong problem on each edge $e \in \mathcal{E}$:

$$\frac{\partial u_e}{\partial t} - \frac{\partial}{\partial s} \left(\mu_e \frac{\partial u_e}{\partial s} \right) + r_e u_e = f_e, \quad s \in (0, \ell_e), \quad t \in (0, T], \quad (B.1)$$

supplemented by the Kirchhoff conditions (10)–(11), the Neumann constraints at the terminal vertices and the global continuity requirements.

Let $V = H^1(\Gamma)$ be the space of functions $w : \Gamma \rightarrow \mathbb{R}$ such that $w_e \in H^1(0, \ell_e)$ for each $e \in \mathcal{E}$ and w is continuous at every junction vertex. We multiply (B.1) by a test function $w \in V$, integrate over e and sum over all edges:

$$\sum_{e \in \mathcal{E}} \int_e \frac{\partial u_e}{\partial t} w ds - \sum_{e \in \mathcal{E}} \int_e \frac{\partial}{\partial s} \left(\mu_e \frac{\partial u_e}{\partial s} \right) w ds + \sum_{e \in \mathcal{E}} \int_e r_e u_e w ds = \sum_{e \in \mathcal{E}} \int_e f_e w ds. \quad (B.2)$$

We integrate by parts the diffusion term on each edge:

$$- \int_e \frac{\partial}{\partial s} \left(\mu_e \frac{\partial u_e}{\partial s} \right) w ds = \int_e \mu_e \frac{\partial u_e}{\partial s} \frac{\partial w}{\partial s} ds - \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e} w \right]_{\partial e}, \quad (B.3)$$

where the boundary term is evaluated at the two endpoints of e with outward normal sign. Summing over all edges and grouping boundary terms by vertex:

$$\sum_{e \in \mathcal{E}} \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e} w \right]_{\partial e} = \sum_{v \in \mathcal{V}} w(v) \sum_{e \in \mathcal{E}_v} \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e} \right]_v. \quad (\text{B.4})$$

We now apply the boundary and junction conditions at each vertex:

- At terminal vertices $v \in \mathcal{T}$: if homogeneous Neumann, $[\mu_e \partial_{\mathbf{n}_e} u_e]_v = 0$ and the contribution vanishes. If Dirichlet, $w(v) = 0$ for test functions in the appropriate subspace and again the contribution vanishes.
- At simple junction vertices, the Kirchhoff condition (10) gives $\sum_{e \in \mathcal{E}_{v_p}} [\mu_e \partial_{\mathbf{n}_e} u_e]_{v_p} = 0$, so the contribution vanishes.
- At generalized junction vertices $v_q \in \mathcal{V}_2$: the Kirchhoff condition (11) gives

$$\sum_{e \in \mathcal{E}_{v_q}} \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e} \right]_{v_q} = c_q \frac{du_q}{dt} + r_q u_q - f_q.$$

Therefore (B.4) becomes:

$$\sum_{e \in \mathcal{E}} \left[\mu_e \frac{\partial u_e}{\partial \mathbf{n}_e} w \right]_{\partial e} = \sum_{v_q \in \mathcal{V}_2} w_q \left(c_q \frac{du_q}{dt} + r_q u_q - f_q \right). \quad (\text{B.5})$$

Substituting (B.3) and (B.5) into (B.2) and rearranging:

$$\sum_{e \in \mathcal{E}} \int_e \frac{\partial u_e}{\partial t} w \, ds + a(u, w) - \sum_{v_q \in \mathcal{V}_2} c_q \frac{du_q}{dt} w_q = F(w) \quad \forall w \in V, \quad (\text{B.6})$$

where

$$a(u, w) = \sum_{e \in \mathcal{E}} \int_e \left[\mu_e \frac{\partial u_e}{\partial s} \frac{\partial w_e}{\partial s} + r_e u_e w_e \right] ds - \sum_{v_q \in \mathcal{V}_2} r_q u_q w_q,$$

$$F(w) = \sum_{e \in \mathcal{E}} \int_e f_e w \, ds - \sum_{v_q \in \mathcal{V}_2} f_q w_q.$$

This is the weak formulation (13).

B.4 Galerkin formulation

Let $V_h \subset V$ be the finite-dimensional subspace spanned by the basis functions $\{\phi_j^e\}_{e \in \mathcal{E}, j=1, \dots, n_e-1}$ (interior node hat functions on each edge) and $\{\phi_v\}_{v \in \mathcal{V}}$ (vertex basis functions), satisfying $\phi_j^e(s_k^{e'}) = \delta_{ee'} \delta_{jk}$ and $\phi_v(v') = \delta_{vv'}$. The discrete solution is written as

$$u_h(s, t) = \sum_{e \in \mathcal{E}} \sum_{j=1}^{n_e-1} \alpha_j^e(t) \phi_j^e(s) + \sum_{v \in \mathcal{V}} \alpha_v(t) \phi_v(s). \quad (\text{B.7})$$

Substituting (B.7) into (B.6) and testing on each basis function, we obtain two families of equations.

Testing on $\phi_k^{e'}$, interior node of edge e' . Since $\phi_k^{e'}$ has support only on e' and vanishes at all vertices, the $\mathcal{V}_2$ term in (B.6) does not contribute. The time term gives:

$$\sum_{e \in \mathcal{E}} \int_e \frac{\partial u_h}{\partial t} \phi_k^{e'} ds = \sum_{j=1}^{n_{e'}-1} \dot{\alpha}_j^{e'} \underbrace{\int_{e'} \phi_j^{e'} \phi_k^{e'} ds}_{P_{kj}^{e'}} + \sum_{v \in \mathcal{V}} \dot{\alpha}_v \underbrace{\int_{e'} \phi_v \phi_k^{e'} ds}_{R_{kv}^{e'}},$$

where only the $e = e'$ term survives due to the local support of $\phi_k^{e'}$ and $R_{kv}^{e'}$ is nonzero only for the (at most two) vertices incident to e' . The stiffness term gives:

$$\begin{aligned} a(u_h, \phi_k^{e'}) &= \sum_{j=1}^{n_{e'}-1} \alpha_j^{e'} \underbrace{\int_{e'} \left[\mu_{e'} \frac{\partial \phi_j^{e'}}{\partial s} \frac{\partial \phi_k^{e'}}{\partial s} + r_{e'} \phi_j^{e'} \phi_k^{e'} \right] ds}_{A_{kj}^{e'}} + \\ &+ \sum_{v \in \mathcal{V}} \alpha_v \underbrace{\int_{e'} \left[\mu_{e'} \frac{\partial \phi_v}{\partial s} \frac{\partial \phi_k^{e'}}{\partial s} + r_{e'} \phi_v \phi_k^{e'} \right] ds}_{B_{kv}^{e'}}. \end{aligned}$$

Testing on $\phi_{v'}$, vertex basis function. Since $\phi_{v'}(v_q) = \delta_{v'v_q}$, the $\mathcal{V}_2$ time term in (B.6) contributes $-c_{v'} \dot{\alpha}_{v'}$ if $v' \in \mathcal{V}_2$ and zero otherwise. The remaining time term gives:

$$\sum_{e \in \mathcal{E}} \int_e \frac{\partial u_h}{\partial t} \phi_{v'} ds = \sum_{e \in \mathcal{E}} \sum_{j=1}^{n_e-1} \dot{\alpha}_j^e \underbrace{\int_e \phi_j^e \phi_{v'} ds}_{R_{v'j}^e} + \sum_{v \in \mathcal{V}} \dot{\alpha}_v \underbrace{\sum_{e \in \mathcal{E}} \int_e \phi_v \phi_{v'} ds}_{Q_{v'v}},$$

where $Q_{vv'}$ is nonzero only when v and v' share at least one edge. The stiffness term gives:

$$\begin{aligned} a(u_h, \phi_{v'}) &= \sum_{e \in \mathcal{E}} \sum_{j=1}^{n_e-1} \alpha_j^e \underbrace{\int_e \left[\mu_e \frac{\partial \phi_j^e}{\partial s} \frac{\partial \phi_{v'}}{\partial s} + r_e \phi_j^e \phi_{v'} \right] ds}_{B_{v'j}^e} + \\ &+ \sum_{v \in \mathcal{V}} \alpha_v \underbrace{\left[\sum_{e \in \mathcal{E}} \int_e \left(\mu_e \frac{\partial \phi_v}{\partial s} \frac{\partial \phi_{v'}}{\partial s} + r_e \phi_v \phi_{v'} \right) ds - r_v \delta_{vv'} \cdot \mathbf{1}_{[v \in \mathcal{V}_2]} \right]}_{G_{v'v}}, \end{aligned}$$

where $\mathbf{1}_{[v \in \mathcal{V}_2]}$ denotes the indicator function of the set $\mathcal{V}_2$, equal to 1 if v is a junction of the second type and 0 otherwise.

B.5 Matrix formulation

Collecting all unknowns into $\mathbf{u} = [\mathbf{u}_{\mathcal{E}}, \mathbf{u}_{\mathcal{V}}]^\top$, where $\mathbf{u}_{\mathcal{E}} = \{\alpha_j^e\}$ and $\mathbf{u}_{\mathcal{V}} = \{\alpha_v\}$, the Galerkin system is

$$\mathbf{M} \dot{\mathbf{u}}(t) + \mathbf{L} \mathbf{u}(t) = \mathbf{f}(t), \quad (\text{B.8})$$

with mass matrix and stiffness matrix

$$\mathbf{M} = \begin{bmatrix} \mathbf{P} & \mathbf{R}^\top \\ \mathbf{R} & \mathbf{Q} + \mathbf{C} \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^\top & \mathbf{G} \end{bmatrix}, \quad (\text{B.9})$$

where the blocks are defined as follows:

- $P_{kj}^e = \int_e \phi_j^e \phi_k^e \, ds;$
- $R_{kv}^e = \int_e \phi_v \phi_k^e \, ds;$
- $Q_{v'v} = \sum_{e \in \mathcal{E}} \int_e \phi_v \phi_{v'} \, ds;$
- $C_{qq} = -c_q$ for $v_q \in \mathcal{V}_2$, zero elsewhere;
- $A_{kj}^e = \int_e \left[\mu_e \frac{\partial \phi_j^e}{\partial s} \frac{\partial \phi_k^e}{\partial s} + r_e \phi_j^e \phi_k^e \right] \, ds;$
- $B_{vj}^e = \int_e \left[\mu_e \frac{\partial \phi_j^e}{\partial s} \frac{\partial \phi_v}{\partial s} + r_e \phi_j^e \phi_v \right] \, ds;$
- $G_{v'v} = \sum_{e \in \mathcal{E}} \int_e \left[\mu_e \frac{\partial \phi_v}{\partial s} \frac{\partial \phi_{v'}}{\partial s} + r_e \phi_v \phi_{v'} \right] \, ds - r_v \delta_{vv'} \cdot \mathbf{1}_{[v \in \mathcal{V}_2]}.$

B.6 Fully discrete system

Discretizing (B.8) in time by the θ -method with uniform step $\Delta t = T/N_T$ and setting $\mathbf{u}^n \approx \mathbf{u}(t^n)$:

$$\mathbf{M} \frac{\mathbf{u}^n - \mathbf{u}^{n-1}}{\Delta t} + \mathbf{L} \left(\theta \mathbf{u}^n + (1 - \theta) \mathbf{u}^{n-1} \right) = \theta \mathbf{f}^n + (1 - \theta) \mathbf{f}^{n-1},$$

which rearranges to

$$\mathbf{H}_\theta \mathbf{u}^n = (\mathbf{M} - (1 - \theta) \Delta t \mathbf{L}) \mathbf{u}^{n-1} + \Delta t \left(\theta \mathbf{f}^n + (1 - \theta) \mathbf{f}^{n-1} \right), \quad (\text{B.10})$$

where $\mathbf{H}_\theta = \mathbf{M} + \theta \Delta t \mathbf{L}$. The choice $\theta = 1$ recovers the implicit Euler method, while $\theta = 1/2$ gives the Crank-Nicolson scheme. Explicitly:

$$\mathbf{H}_\theta = \begin{bmatrix} \mathbf{P} + \theta \Delta t \mathbf{A} & \mathbf{R}^\top + \theta \Delta t \mathbf{B} \\ \mathbf{R} + \theta \Delta t \mathbf{B}^\top & \mathbf{Q} + \mathbf{C} + \theta \Delta t \mathbf{G} \end{bmatrix}. \quad (\text{B.11})$$

Since $a(\cdot, \cdot)$ is symmetric and coercive on V and $\mathbf{M}$ is symmetric positive definite, $\mathbf{H}_\theta$ is symmetric positive definite for any $\Delta t > 0$ and $\theta \in (0, 1]$ and the algebraic formulation of the parabolic case of partial differential equations on graphs admits a unique solution at each time step.

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